



Book of Abstracts

Alchemy Workshop in Free Energy Methods for Drug Design

*May 4-6
Barcelona*

WELCOME

The Free Energy Workshop is a recurring event hosted by the Alchemistry Organization Committee and the Open Molecular Software Foundation (OMSF). Since 2018, it has served as a key meeting point for the international community working on free energy methods in computational chemistry and drug design, bringing together researchers to share advances, exchange ideas, and foster collaboration across the field.

This 2026 edition marks third time the workshop is held in Europe, and we are very pleased to organize it in Barcelona for the first time. The event continues to grow as a central forum for the community, welcoming hundreds of participants from around the world each year.

The main objective of this year's workshop is to focus on the current challenges in the field, with the aim of helping to define the next directions, opportunities, and methodological developments in free energy calculations and their applications.

We would also like to sincerely thank our sponsors for their support in making this event possible.

We are delighted to welcome all participants and contributors, and we hope this collection of abstracts provides a useful overview of the exciting work being presented throughout the workshop.



2026 ORGANIZING COMMITTEE

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Aakanshit (AK) Nandkeolyar - Ph.D. Candidate, University of California Irvine

4th May 2026

5th May 2026

6th May 2026

8:30 9:00	Attendee registration Welcome + opening remarks
Session 1:	
Advanced Sampling & Algorithmic Innovations Chairs: Anna H Herz, Carolina Estarellas	
9:15	Francesco Luigi Gervasio When Collective Variables Fail: Innovative Solutions for Converging Free Energy Landscapes
9:40	Domen Pregeljc Are Multiscale Simulations More Accurate Than Classical Alternatives? An Application to Atropisomerism and Development of a Multistate QM/ML/MM Free-Energy Method
10:05	Avi Robinson-Mosher Times Square sampling: a guided tour of an adaptive algorithm for free energy estimation
10:30	Break
Session 2:	
Advanced Sampling & Algorithmic Innovations Chairs: Aleix Quintana, Carolina Estarellas	
11:00	Justina Ratkeviciute Improving Alchemical Binding Free Energy Calculations Using Fully Adaptive Simulated Tempering (FAST)
11:25	Adam Baskerville Binding Free Energies from Following Ligands on the Way Out
11:50	Narjes Ansari Lambda-ABF-OPES: A Robust Framework for Absolute and Relative Binding Free Energy Calculations
12:15	Flash poster presentations
12:40	Lunch
Session 3:	
Integrating Physics-Based Modeling & Machine Learning Chairs: Ernestas Urniezius, Jenke Scheen	
14:10	Patrick Walters AI in Drug Discovery - Revolution, Evolution, or Complete Nonsense?
15:00	Haoyu Yu Learning the Thermodynamic Landscape of Protein-Ligand Interactions at Scale
15:25	Marco Cecchini On the impact of AI technologies on virtual screening for drug discovery
15:50	Break
Session 4:	
Integrating Physics-Based Modeling & Machine Learning Chairs: Emily Landwehr, Jenke Scheen	
16:20	Márton Vass No Structure? No Problem: Just Have Some TEA with FEPI
16:45	Michael Shirts Mapping between Boltzmann distributions: from analytical to machine-learned mappings
17:10	Jasmin Guven Methods for alchemical free energy calculations with betalactamase inhibitors
17:35	Poster session
19:00	End of the Day
19:30	Guided City Tour

9:00	Attendee registration + Walk in
Session 5:	
Proteins, Peptides & Novel Modalities Chairs: Finlay Clark, Jordi Juárez-Jiménez	
9:00	Julien Michel Beyond small molecule-protein complexes: relative binding free energy calculations for diverse modalities
9:25	Luca Carlini Dissociation Free Energy (DFE) Methodology for Identification of Novel Interfaces in Molecular Glue Applications
9:50	Patricia Blanco Gabella Understanding the Role of H-Bonds in the Stability of Molecular Glue-Induced Ternary Complexes
10:15	Break
Session 6:	
Proteins, Peptides & Novel Modalities Chairs: Sarah Slietitz, Jordi Juárez-Jiménez	
10:45	Raquel López-Ríos de Castro Prediction of Protein-Protein binding free energy upon mutations using ML/G
11:10	Hugo Gutiérrez de Terán Accurate predictions of protein mutational effects with QresFEP
11:35	Laura Perez Benito Large scale FEP for peptides
12:00	Flash poster presentations
12:25	Group picture
12:40	Lunch
Session 7:	
High-Throughput FEP & Scalable Workflows Chairs: Yarbina Ivanova, Dmitry Lupyán	
14:10	Jonathan Essex Grand Canonical Simulation Methods in Structure-Based Drug Design
15:00	Joshua Horton Recent Advances in OpenFE: From Workflows to Multi-Method Alchemical Networks
15:25	Willem Jespers Doing More with Less: Accurate and Scalable Ligand Free Energy Calculations by Focusing on the Binding Site
15:50	Break
Session 8:	
High-Throughput FEP & Scalable Workflows Chairs: Chenggong Hui, Dmitry Lupyán	
16:20	Mark Mackey Low-cost high-throughput FEP using active learning with 3D features
16:45	Forrest York High Throughput FEP With Local Sampling
17:10	Robert Abel Advancing Candidate Optimization via Rigorous Free Energy Calculations
17:35	Poster session
19:00	End of the Day
19:30	Happy Hour by Schrödinger

9:00	Attendee registration + Walk in
Session 9:	
Industrial Applications & Development Chairs: Alzbeta Kubincova, Salomé Liabrés	
9:00	Joseph Black The application of free energy methods at Bayer
9:25	Stephan Ehrlich From pilot to practice: Embedding FEP into DMTA cycles for prospective decision-making at UCB
9:50	David Pearfman Addressing Electronic Complexity in FEP: Scaling QMMM and Physics-Informed ML for Predictive Lead Optimization
10:15	Kenneth Goossens Physics-based and LQM-Driven Drug Discovery Applied to Neurodegeneration
10:40	Break
Session 10:	
High-Throughput FEP & Scalable Workflows Chairs: Emily Landwehr, Salomé Liabrés	
11:10	Vincent Voelz Improved expanded ensemble methods for massively parallel virtual screening
11:35	Ramon Crehuet Introducing NEMAT, a non-equilibrium method to calculate binding energies to membrane proteins
12:00	Jonah Vilecek Scaling LaDyBUGS for Drug Discovery: Methodological Enhancements and Expanded Protein-Ligand Benchmarks
12:25	Chris Neale Chimeric Atom Matching and Sampling Sufficiency as Determinants of RBFE Accuracy
12:40	Final remarks
12:45	Lunch
14:00	End of the Day



Guided city tour

Starting point:
Entrance of the
campus UPF Ciutadella

19:30

Happy Hour by Schrödinger



"Blue Terrace"

Sofitel Barcelona Skipper
Av. del Litoral, 10, Ciutat Vella

19:00

KEYNOTE LECTURE

K1 - AI in Drug Discovery - Revolution, Evolution, or Complete Nonsense?

Patrick Walters

OpenADMET

K2 - Grand Canonical Simulations for In Silico Prediction of Fragment Binding Sites, Modes, and Affinities

Jonathan Essex, William Poole, Victoria Nathan-Maister, Oliver Melling, Marley Samways, Davide Branduardi, Richard Taylor, Marcel Verdonk

University of Southampton

Fragment based drug discovery (FBDD) is widely used in the pharmaceutical industry as a route to generating lead compounds. Through knowledge of their binding sites, fragment hits may be combined into single molecules with good potency and physical properties. Computational FBDD supports experiment by providing a route to library design, virtual screening, binding site identification and binding affinity prediction.

In this talk, the development and application of grand canonical non-equilibrium Monte Carlo (GCNEMC) in this domain will be discussed. By combining non-equilibrium move proposals, with grand canonical Monte Carlo acceptance tests, we are able to insert and delete small molecules into the binding sites of host-guest and protein systems, much more efficiently than conventional molecular dynamics. Through these simulations we are able to identify potential ligand binding sites by augmenting the sampling in mixed-solvent molecular dynamics. Fragment binding poses, including situations where multiple binding poses have been reported, are also readily identified. Finally, by varying the chemical potential of the simulations, absolute ligand binding free energies may be calculated, without the need for restraints or corrections to address multiple binding poses. Binding sites, poses, and affinities may all be calculated through a single series of simulations run at different chemical potentials. Finally, cases where methodological improvements are needed will be discussed.

ORAL PRESENTATIONS

Session 1: Advanced Sampling & Algorithmic Innovations

01 - When Collective Variable Fail: Innovative Solutions for Converging Free Energy Landscapes

Francesco Luigi Gervasio

School of Pharmaceutical Sciences and Institute of Pharmaceutical Sciences of Western Switzerland (ISPSO), University of Geneva

Many enhanced sampling techniques rely on the definition of collective variables (CVs) to effectively explore free energy landscapes. However, defining a set of optimal CVs that approximate the reaction coordinate of complex biomolecular systems is far from trivial, even for seemingly trivial cases such as the binding of a ligand to a protein. Using sub-optimal CVs can lead to free energies that are not converged. In the past, we have developed various approaches based on multiple replicas [1] or path-like collective variables, both to explore complex free energy landscapes and to compute their free energy profiles [2]. However, this and similar approaches face a number of limitations. In this talk I will show two recent development from my group that try and address the problem of converging free energy landscapes with sub-optimal CVs. The first is OneOPES, a multiple replica approach that greatly ameliorates the problem of free energy convergence with sub-optimal CVs [3]. The other is a new path-like collective variable called the 'Deep-locally-non-linear-embedding', which is inspired by principles of the locally linear embedding technique and is trained on a reactive trajectory [4]. I will show the effectiveness of these methods in ligand binding, receptor activation and protein folding [5-7].

1. G Bussi, et al JACS 128, 13435-13441 (2006)
2. Branduardi, et al. JCP. 126, 054103 (2007)
3. Rizzi, et al JCTC 19, 5731–5742 (2023)
4. Fröhling, et al. JCP 160, 174109 (2024)
5. Febrer et al. JCTC J. 20, 1275 (2024).
6. Aureli et al. JCTC 21, 7687–7700 (2025)
7. Rizzi et al JCPL 16, 9636–9645 (2025).

O2 - Are Multiscale Simulations More Accurate Than Classical Alternatives? An Application to Atropisomerism and Development of a Multistate QM(ML)/MM Free-Energy Method

Domen Pregeljč, Sereina Riniker

ETH Zurich

Multiscale QM/MM molecular dynamics (MD) simulations should in principle represent a more precise description of the system compared to their classical counterparts. In the present work, we put this hypothesis to the test by assessing the ability of QM/MM and classical MD simulations to capture subtle stereoelectronic effects. A series of molecular balances is studied and the results are compared to experiment. While the relative free-energy differences calculated are similar, the obtained potentials of mean force differ significantly between QM/MM and MM. A detailed geometric analysis is carried out and the force field terms giving rise to the observed differences are highlighted. The molecular balances are a good example where the approximations made in a classical force field are inadequate. To address this issue in free-energy calculation, we have developed a multiscale multistate method. We will discuss how QM/MM treatment has been integrated into the replica-exchange enveloping distribution sampling (RE-EDS) free-energy method, thereby allowing for efficient and accurate free-energy calculations from a single MD simulation. Several semi-empirical QM methods have been tested and the results on hydration free energies were compared to classical force fields and experiment. The effect of the water model and the importance of QM-MM model compatibility were also studied. Furthermore, as the introduced framework is QM-model agnostic, a machine-learned interatomic potential (MLIP) can be incorporated to substitute the semi-empirical treatment for better accuracy but feasible computational cost.

O3 - Times Square sampling: a guided tour of an adaptive algorithm for free energy estimation

Avi Robinson-Mosher, Cristian Predescu, Michael Snarski, Duluxan Sritharan, Tamas Szalay, and David E. Shaw

D. E. Shaw Research

Estimating free energy differences, an important problem in computational drug discovery and in a wide range of other application areas, commonly involves a computationally intensive process of sampling a family of high-dimensional probability distributions and a procedure for computing estimates based on those samples. The variance of the free energy estimate of interest typically depends strongly on how the total computational resources available for sampling are divided among the distributions, but determining an efficient allocation is difficult without sampling the distributions. In this talk I will discuss the Times Square sampling algorithm, an on-the-fly estimation method that dynamically allocates resources in such a way as to significantly accelerate the estimation of free energies and other observables, while providing rigorous convergence guarantees for the estimators.

Session 2: Advanced Sampling & Algorithmic Innovations

04 - Improving Alchemical Binding Free Energy Calculations Using Fully Adaptive Simulated Tempering (FAST)

Justina Ratkeviciute, Miroslav Suruzhon, Khaled Abdel-Maksoud, Anna Cavalleri, Michael S. Bodnarchuk, Ian D. Wall, Marc Daemgen, Jonathan W. Essex

University of Southampton

Alchemical binding free energy (AFE) calculations are often performed at relatively short timescales, during which many ligand binding modes may not be sufficiently sampled due to high kinetic barriers. As a result, these calculations are often highly dependent on the initial structural choices, and enhanced sampling methods are needed to ease this bias. An additional challenge with AFE calculations is designing an optimal protocol of intermediate states, which can often lead to poor efficiency and a trade-off between sufficient long-timescale sampling and adequate convergence.

In this work we present fully adaptive simulated tempering (FAST) – a novel and robust variation of the simulated tempering and sequential Monte Carlo algorithms that estimates both the free energy profile and the optimal interpolation protocol on-the-fly without the need for system-specific knowledge. Alongside improving efficiency in traversing the intermediate states, this method also achieves increased effective decorrelation from the initial coordinates as the entire simulation time is spent on a single, continuous trajectory. As a result, FAST addresses both previously described AFE challenges.

This algorithm can be used in hydration and binding free energy calculations, enhanced sampling of binding modes, and any Markov chains combining the two. In this work we will demonstrate how FAST deals with a variety of systems, from simple solutes to more challenging protein-ligand systems such as p38 and HSP90, showcasing its sampling efficiency and quality of free energy estimates in comparison with standard AFE calculations.

05 - Binding Free Energies from Following Ligands on the Way Out

Adam Baskerville

Kvantify

Accurate prediction of protein-ligand binding free energies remains a continued challenge in computational drug discovery. Alchemical approaches such as free energy perturbation (FEP) can achieve high accuracy but are often too computationally expensive for routine use in lead optimisation, while approximate scoring functions lack rigorous physical foundations.

Here, we present a fully physics-based workflow for estimating binding free energies using umbrella sampling (US) to construct the potential of mean force along ligand unbinding pathways. A common limitation of traditional US is the definition of an appropriate reaction coordinate; in this work, it is obtained directly from physically realistic unbinding trajectories generated using Koffee, our fast, proprietary kinetics engine [1].

By sampling along a physically motivated unbinding trajectory, the method focuses sampling on free energy relevant degrees of freedom, avoiding inefficiencies associated with unphysical or weakly overlapping intermediate states while retaining rigorous statistical sampling. This eliminates reliance on empirical scoring functions or machine-learned models and reduces the computational cost relative to conventional absolute binding free energy calculations. Furthermore, unlike alchemical transformations, this physical pathway approach yields valuable mechanistic insights, including the characterisation of transition states that are inaccessible to non-physical perturbation methods.

We will outline the theoretical framework of the approach and present early results demonstrating its efficacy across both congeneric series and diverse non-congeneric scaffolds where traditional relative free energy methods often struggle.

[1] Madsen, N. Kristian et al., 2025, Unlocking Scalable Ligand Residence Time Predictions with Koffee Unbinding Kinetics Simulations, bioRxiv, 2025., <http://biorxiv.org/content/early/2025/12/02/2025.11.06.686759.abstract>, <https://www.kvantify.com/products/kinetics>.

O6 - Lambda-ABF-OPES: A Robust Framework for Absolute and Relative Binding Free Energy Calculations

Narjes Ansari, Félix Aviat, Jérôme Hénin, ean-Philip Piquemal, Louis Lagardère

Qubit Pharmaceuticals

The accurate prediction of protein-ligand binding free energies remains a computational bottleneck in drug discovery. This work presents a unified advancement in alchemical simulations through the development of Lambda-ABF-OPES and its dual-topology derivative, dual-LAO. By coupling adaptive biasing forces with on-the-fly probability enhanced sampling and the AMOEBA polarizable force field, these methods achieve significant accelerations—ranging from 9-fold for absolute binding energies to 30-fold for relative transformations. Together, they enable the rapid, systematic, and accurate modeling of both simple and complex molecular changes, providing a robust framework for high-throughput lead optimization and rational drug design.

Session 3: Integrating Physics-Based Modeling & Machine Learning

07 - Learning the Thermodynamic Landscape of Protein-Ligand Interactions at Scale

Haoyu Yu, Yusong Wang, Youjun Xu, Shaoning Li, Qilong Wu, Qiaojing Huang, Wenjuan Tan, Wentao Li, Nanjun Chen

Anew Therapeutics

While AlphaFold3 (AF3) has revolutionized static structure prediction, biological function is inherently rooted in dynamic conformational ensembles. Diffusion methods employed in AF3 struggle to capture these thermodynamic distributions, stemming from a mathematical incompatibility between heuristic alignment training and generative sampling. Here, we introduce AnewMD, a framework that repurposes AF3 into a rigorous thermodynamic sampler, bypassing the need for traditional molecular dynamics (MD) simulations to recover conformational ensembles. By implementing quotient-space diffusion with a closed-form $SE(3)$ projection, we resolve alignment-sampling inconsistencies to guarantee the faithful recovery of the Boltzmann distribution. Enhanced by ergodic template guidance and sub-Angstrom geometric supervision, AnewMD generates MD-quality thermodynamic distributions instantaneously, eliminating the requirement for expensive, time-step-based numerical integration. As the first transferable, all-atom generative surrogate, AnewMD unlocks scalable, dynamic structure-based drug design without the computational wall-clock of MD.

O8 - On the impact of AI technologies on virtual screening for drug discovery

Marco Cecchini and Hryhory Sinenka

University of Strasbourg

AI foundational models for predicting protein-ligand interactions and binding affinities have recently started to emerge. We challenged one of them, i.e. Boltz-2, on a difficult dataset constructed on ten ultra-large virtual screening hit lists of pharmacologically relevant targets with in vitro binding assays. We show that Boltz-2 is the best classifier, with a success rate twice that of any other fast rescoring strategy. Ligand classifications by Boltz-2 are straightforward, accurate, efficient and robust, opening to million-compound accurate rankings on mid-sized GPU clusters. Based on the accuracy, robustness, and output rate, AI foundational models like Boltz-2 appear to fill a critical gap between ultra-large library docking and lead optimisation in a novel and elegant fashion.

Session 4: Integrating Physics-Based Modeling & Machine Learning

O9 - No Structure? No Problem: Just Have Some TEA with FEP!

Márton Vass, Griffin Young, Viraj Jayaram, Liwei Chang

Schrödinger Technologies Ltd

Obtaining accurate FEP models for experimentally determined protein-ligand complexes is generally easily achievable, however, when there are no available structures with the chemical series of interest, or even no structures of the protein and its homologs, this task is markedly more difficult. Schrödinger's Target Enablement Accelerator (TEA) provides significant speedups to such novel target enablement workflows. First it automatically collects compound SAR and protein homologs, and builds template protein-ligand complexes using automated homology modelling, ML protein folding, docking, and ML co-folding methods. Then it refines the templates using induced fit docking with MD simulations (IFD-MD) and creates ready-to-run FEP maps. Finally it runs automated FEP model selection using various relative and absolute binding FEP methods. Our approach provides users expert tools to rationalise the construction of accurate ligand potency models, rather than relying on a black box model. This streamlined approach allows researchers to rapidly establish predictive models for previously undrugged or uncharacterised targets. So you have time for a cup of tea while running TEA!

O10 - Mapping between Boltzmann distributions: from analytical to machine-learned mappings

Michael Shirts, Aidan Wegner, Edgar Olehnovics, Matteo Salvalaglio

University of Colorado Boulder

We review the theory behind mapping between conformational distributions, show how even simple mappings can accelerate the convergence of thermodynamic averages and free energy differences, and review how machine learning methods can be used to learn such mappings. We specifically show how these techniques can be used to calculate free energy differences between realistic crystal polymorphs relevant to drug design without using alchemical intermediates, instead using configuration space maps learned with normalizing flow techniques. We explore how physical insights about the systems of interest can improve our ability to construct even machine-learned maps between physical states.

O11 - Towards improved binding affinity calculations for metalloproteins with meze

Jasmin Güven, Papu Kalita, Marko Hanzževački, Philip Hinchcliffe, Jim Spencer, Antonia Mey and Adrian Mulholland

University of Bristol

Metalloproteins are abundant in nature and play key roles in enzyme catalysis and drug discovery. However, computational methods for accurately modelling these systems are limited and difficult to set up. Existing tools also lack direct interfaces with open-source workflows for relative binding free energy (RBFE) calculations. Furthermore, RBFE methods have not been extensively evaluated with such systems.

Here, we present meze, a Python-based metalloenzyme parameterisation program designed to simplify the setup of molecular dynamics (MD) simulations and RBFE calculations for small molecules binding to zinc-containing metalloproteins. Built on BioSimSpace and MDAnalysis, meze implements three zinc models: a restraint-based nonbonded approach using flat-bottomed distance and angle restraints, the upgraded Amber force field (UAFF), and a modified extended zinc Amber force field (eZAFF) method. In addition to parameterisation, meze automates system solvation, equilibration, alchemical network preparation, and production simulations.

We applied meze to parameterise a set of 51 indole carboxylate inhibitors binding to two metallo-beta-lactamases (MBLs) for MD simulations and RBFE calculations. One MBL is a dimer, demonstrating the generalisability of meze to different systems. Building on our previous work with 16 heteroaryl phosphonates, here we evaluate two new metal models to improve RBFE accuracy. The hybrid bonded model, which assigns DFT-based partial charges, produces more stable zinc coordination during MD, and is expected to lead to better experimental RBFE agreement than nonbonded approaches. These results provide a foundation for computational inhibitor design against MBLs and highlight improved zinc modelling as a key priority for future development. meze is openly available and, owing to its modular design, can readily be extended to incorporate additional metal modelling approaches.

Session 5: Proteins, Peptides & Novel Modalities

O12 - Beyond small molecule–protein complexes: relative binding free energy calculations for diverse modalities

Julien Michel

OpenBioSim & University of Edinburgh

A broad range of solutions is now available to practitioners seeking to perform Relative Binding Free Energy (RBFE) calculations on complexes between proteins and drug-like small molecules. This presentation will focus on emerging free energy methodologies and associated open-source software implementations that combine components developed at the University of Edinburgh, OpenBioSim, and by the broader community, with the aim of expanding the applicability of alchemical free energy calculations to challenging drug targets and modalities.

These advances will be illustrated through case studies involving bespoke RBFE protocols to:

- i) profile the drug-resistance landscape of p53/MDM2 inhibitors against protein side-chain mutations in an intrinsically disordered region of MDM2;*
- ii) model the binding of reversible covalent inhibitors of fibroblast activation protein;*
- iii) model complex ring-breaking and linker transformations within a dataset of macrocyclic ligands targeting PCSK9.*

Throughout the presentation, emphasis will be placed on methodological considerations required to achieve accurate and converged binding free energy estimates, before concluding with a discussion of open questions the field must address to further broaden the applicability of RBFE methodologies in drug discovery.

O13 - Dissociation Free Energy (DFE) Methodology for Identification of Novel Interfaces in Molecular Glue Applications

Luca Carlino

AstraZeneca

Modulating protein complexes with molecular glues, such as thalidomide-class immunomodulatory agents that redirect substrate recognition by E3 ligases, aryl-sulfonamides that recruit splicing factors for selective protein turnover, and synthetic stabilisers of 14-3-3 adaptor interactions with client proteins, represents an emerging approach in oncology for targeting cancer cell survival pathways. Correctly predicting the geometries of ternary complexes involving molecular glues is crucial for the design of new small molecules through structure-based approaches. AI-based tools as well as protein-protein docking software have limitations in predicting accurate geometries, often resulting in the desired complexes being lost within the generated output. We have developed a physics-based workflow able to correctly describe the affinity values of a diverse range of ternary glue complexes. The workflow considers all interactions at the complex interface and is based on advanced MD simulations aimed to pull a part the proteins in the ternary complex and derive the dissociation free energy (DFE).¹ To demonstrate the applicability and limitations of the methodology, specific case studies will be presented.

¹Wang et al. Scientific Reports, 2022

O14 - Understanding the Role of H-Bonds in the Stability of Molecular Glue-Induced Ternary Complexes

Patricia Blanco-Gabella; Varbina Ivanova; Álvaro Serrano-Morrás; Julian E. Fuchs; Jordi Juárez-Jiménez

University of Barcelona

Protein–protein interaction (PPI) networks play a central role in many biological processes, and thus, the possibility of modulating them using small molecules offers several therapeutic opportunities. [1,2] Molecular glues (MGs) are small molecules that bind to a PPI interface and stabilize the complex. Oftentimes, MGs show no measurable affinity for at least one of the proteins involved in the ternary complex [3,4], and the molecular bases for their action are not completely understood. We previously reported a significant correlation between protein–protein hydrogen bond robustness and the stability of the CRBN–CK1 α complex induced by the antimyeloma drug lenalidomide. [5]

In this work, we demonstrate that this relationship is not unique for that system but rather represents a reproducible physicochemical phenomenon underlying the mechanism of action of chemically diverse MGs, including additional IMiDs and Fusicoccin A. [6] Our results shed light on a vaguely understood phenomenon and pave the way for the development of new computational methods that enable the rational discovery of molecular glues.

1. Stanton B.Z., *Science*, 359,eaao5902 (2018)
2. Konstantinidou M., *Cell Chem Biol*, 31(6), 1064-1088 (2024)
3. Bussiere D., *Nat Chem Biol*, 16(1), 15-23 (2020)
4. Cao S., *Nat. Commun*, 13(1), 815
5. Miñarro-Lleonar M., *J Med Chem*, 66 (9), 6037-6046 (2023)
6. Blanco-Gabella P., *J. Chem. Inf. Model.*, Accepted (2026)

Session 6: Proteins, Peptides & Novel Modalities

O15 - Prediction of Protein–Protein binding free energy upon mutations using MLCG (Machine-Learned Coarse-Grained Model)

Raquel López-Ríos de Castro, Klara Bonneau, Cecilia Clementi

Department of Physics, Freie Universität Berlin, Arnimallee 12, Berlin, Germany

Protein–protein interactions (PPIs) play a central role in many biological processes, including signal transduction, enzyme regulation, and cellular signaling networks. [1,2] It is estimated that more than 80% of proteins function as part of complexes rather than in isolation. [1] Therefore, predicting and quantifying the effects of mutations on protein–protein binding affinity is essential for understanding molecular recognition, interpreting disease-associated protein variants, and guiding protein engineering and drug design. [1,2] In particular, accurate estimation of changes in binding free energy ($\Delta\Delta G$) upon mutation remains a major challenge in computational biophysics. [3]

State-of-the-art approaches for computing mutation-induced changes in protein-protein binding free energy typically rely on atomistic molecular dynamics simulations combined with enhanced sampling techniques or alchemical free energy calculations. [3] While these methods can achieve high accuracy, they often require complex computational pipelines involving replica exchange or carefully designed alchemical transformations. Moreover, atomistic enhanced sampling simulations are computationally expensive, making them difficult to scale up to high-throughput applications such as large mutational scans or drug discovery workflows.

Here, we develop a computational framework to efficiently estimate mutation-induced changes in protein–protein binding free energy. Our approach combines the machine-learned coarse-grained (MLCG) model developed by the Clementi lab [4] with RANGE to account for long-range interactions. [5] Using this pipeline, we perform short coarse-grained molecular dynamics simulations of both wild-type and mutant protein complexes and estimate binding free energies using the Multistate Bennett Acceptance Ratio (MBAR), allowing the computation of free energy differences between bound and unbound states.

We validate our approach on the barnase–barstar complex, a well-established benchmark system for protein–protein binding studies. Despite using short coarse-grained simulations and avoiding explicit alchemical transformation pathways, our method achieves accuracy comparable to state-of-the-art atomistic simulations, with prediction errors below 1 kcal/mol. We further evaluate the transferability of our approach using a curated set of protein–protein interaction mutations from the SKEMPI database, [6] where our method maintains good predictive performance across multiple systems.

The reduced computational cost of our framework enables automated and scalable calculations, making it well suited for high-throughput applications. This approach provides a promising strategy for studying biomolecular interactions and accelerating computational protein design.

References

- [1] Berggård, Tord, Sara Linse, and Peter James. "Methods for the detection and analysis of protein–protein interactions." *Proteomics* 7.16 (2007): 2833-2842.
- [2] Rao, V. Srinivasa, et al. "Protein-protein interaction detection: methods and analysis. " *International journal of proteomics* 2014.1 (2014): 147648.
- [3] Zhang, Ivy, et al. "Identifying and overcoming the sampling challenges in relative binding free energy calculations of a model protein: protein complex." *Journal of chemical theory and computation* 19.15 (2023): 4863-4882.
- [4] Charron, Nicholas E., et al. "Navigating protein landscapes with a machine-learned transferable coarse-grained model." *Nature chemistry* 17.8 (2025): 1284-1292.
- [5] Caruso, Alessandro, et al. "Extending the range of graph neural networks with global encodings." *Nature Communications* 17.1 (2026): 1855
- [6] Moal, Iain H., and Juan Fernández-Recio. "SKEMPI: a structural kinetic and energetic database of mutant protein interactions and its use in empirical models. " *Bioinformatics* 28.20 (2012): 2600-2607

O16 - Accurate predictions of protein mutational effects with QresFEP

Hugo Gutiérrez de Terán, Lucien Koenekoop, Johan Åqvist

Research Center of Nanomaterials and Nanotechnology / CSIC

I will present QresFEP-2, a novel hybrid-topology FEP protocol benchmarked on a comprehensive protein stability dataset of 10 protein systems, encompassing almost 600 mutation, combining excellent accuracy (average MAE = 1.25 kcal/mol) with probably the highest computational efficiency among available FEP protocols (480 CPU-h/mutation). Case studies include assessing the thermodynamic stability of over 400 mutations generated by a systematic mutation scan of the 56-residue B1 domain of streptococcal protein G (Gβ1), site-directed mutagenesis on protein ligand binding to a prototypical GPCR, and protein-protein interactions. While this dataset was very recently published, we will herein disclose adaptations of the QresFEP-2 protocol to account for charge-changing mutations, where a benchmark on over 200 mutants confirms that the accuracy remains remarkably satisfactory. The dataset of charge-changing mutations also allows to investigate effects of local environment on residue properties, and how these should be considered when performing such calculations.

O17 - Large scale FEP for peptides

Laura Perez Benito

Johnson & Johnson

Peptide modalities have emerged as a powerful and versatile therapeutic class, strategically positioned between small molecules and large biologics. Their unique structural and functional characteristics enable exceptional target specificity, tunable pharmacokinetics, and the capacity to modulate protein–protein interactions and other targets traditionally considered “undruggable.” Accurate binding free energy calculations are therefore critical for advancing peptide candidates from early hits to potent, well-optimized binders. While alchemical free energy perturbation (FEP) methods have become a cornerstone of lead optimization for small molecules, their application to peptides presents significant challenges due to increased molecular complexity, conformational heterogeneity, and demanding system preparation. Here, we describe key obstacles encountered when deploying FEP in peptide discovery programs and introduce a semi-automated workflow designed to streamline peptide preparation and enable large-scale FEP campaigns. We demonstrate the robustness of this approach through both retrospective benchmarks and prospective applications to peptide libraries, highlighting its potential to accelerate optimization of peptide therapeutics. Workflows leveraging Moe and Glide enabled reliable binding pose generation for FEP during library enumeration, including R-group decomposition and constrained docking. Retrospective FEP validation at several positions showed good correlation sufficient to deprioritise inactive and enrich potent peptides. Prospective FEP at key positions identified new moieties with improved potency increasing biochemical potency 10-fold and cellular potency 3-fold.

Session 7: High-Throughput FEP & Scalable Workflows

O18 - Recent Advances in OpenFE: From Workflows to Multi-Method Alchemical Networks

Joshua, Horton, Irfan Alibay, Hannah Baumann, Finlay Clark, Daniel Cole

Open Free Energy (OMSF)

Alchemical free energy methods have found tremendous success over the last few decades, becoming key components of drug discovery pipelines. Despite many and ongoing innovations in this area, with several methodological improvements published each year, it remains challenging to consistently run free energy campaigns using state-of-the-art tools and best practices. In many cases, doing so requires expert knowledge, and/or the use of expensive closed source software. The Open Free Energy project (<https://openfree.energy/>) was created with the aim of addressing these issues. A joint effort between several academic and industry partners, the consortium aims to create and maintain reproducible and extensible open source tools for running large scale free energy campaigns.

In this presentation, we will cover recent developments that expand OpenFE's capabilities across workflows, protocols, and execution. Methodological updates include support for membrane systems within OpenFE protocols, broadening the range of applicable targets. We also introduce multi-method network strategies combining the hybrid topology protocol for closely related ligands with the separated topology protocol for larger or more complex transformations. This allows users to construct networks according to chemical similarity, increasing flexibility and efficiency. We will also highlight recent benchmark efforts evaluating the performance of the OpenFE tooling in production settings.

O19 - Doing More with Less: Accurate and Scalable Ligand Free Energy Calculations by Focusing on the Binding Site

Willem Jespers, David Alencar Araripe, Shen Guo, Donald van Pinxteren, Viktor Prypoten, Michiel Jespers, Daan Jiskoot, Linde Schoenmaker, Johan Åqvist, Hugo Gutiérrez-de-Terán

Groningen University

Accurate and computationally efficient prediction of protein–ligand binding free energies is a key objective in structure-based drug discovery. Free energy perturbation (FEP) methods can provide high predictive accuracy, but their broader adoption is often limited by sampling cost, scalability, and hardware requirements. In this presentation, I describe recent methodological and infrastructural advances implemented in the QligFEP framework that address these challenges.

I first summarize recent benchmark results demonstrating that reliable relative binding free energies can be obtained by focusing sampling on the binding site while maintaining agreement with experimental data. This work has been extended to include protein–ligand systems involving membrane proteins, while still enabling simulations with a comparable system size of approximately 6,500 atoms. Performance and robustness are evaluated across diverse chemical transformations relevant to lead optimization.

In addition to equilibrium FEP calculations, I present a non-equilibrium free energy methodology based on fast switching simulations. This approach enables rapid estimation of relative binding free energies and offers a favorable balance between computational cost and accuracy. I discuss practical trade-offs between equilibrium and non-equilibrium protocols and illustrate use cases where non-equilibrium methods significantly reduce time-to-result while retaining predictive power.

To enable large-scale and flexible deployment, I introduce a new cloud-based infrastructure for running QligFEP workflows on demand, supporting both CPU and GPU resources. I also present the latest iteration of an ARM-based Raspberry Pi cluster, highlighting its role as a low-cost, energy-efficient platform for high-throughput testing, method development, and education. Finally, I report performance benchmarks of QligFEP on GPU architectures, demonstrating substantial speedups relative to CPU-only execution.

Together, these developments establish QligFEP as a versatile platform for equilibrium and non-equilibrium free energy calculations across diverse molecular systems and computing environments.

Session 8: High-Throughput FEP & Scalable Workflows

O20 - Low-cost high-throughput FEP using active learning with 3D features

Mark Mackey

Cresset

We present an active learning cycle combining AI/ML with free energy calculations that can locate the majority of the high-binding molecules in a pool of suggestions while performing free energy calculations only on a small fraction. A number of retrospective test cases will be presented, along with some prospective successes.

O21 - High Throughput FEP With Local Sampling

Forrest York

Glysade

RBFE simulations are accurate but slow and costly, limiting their use at scale. We show how local sampling can accelerate rigorous RBFE calculations to <15 minutes per edge without compromising accuracy. We compare this approach to global sampling and highlight implications for faster, cheaper, and broader RBFE-powered design.

022 - Advancing Candidate Optimization via Rigorous Free Energy Calculations: From Kinome-Wide Selectivity to Solid-State Properties

Robert Abel

Schrodinger

The efficient identification of high-quality development candidates remains hindered by late-stage failures due to unforeseen toxicity or suboptimal physicochemical properties. To address this, we are integrating a suite of advanced modeling tools designed to accelerate the design-make-test-analyze cycle by centering free energy calculations as the primary engine for property prediction.

Central to this effort is the development of structure-based in silico selectivity panels, conceived as computational analogues to experimental assays. We present results focused on kinome-wide selectivity, delivering predictive accuracy comparable to standard in vitro panels (e.g., ScanMAX) while providing deeper mechanistic insights into the underlying thermodynamics of binding. This framework is currently being expanded to encompass broader off-target and toxicity endpoints.

Beyond selectivity, we detail the application of free energy calculations to predict small molecule crystal structure polymorphs and their associated properties, including the estimation of polymorph-specific solubility crucial for formulation and bioavailability. These predictive models are leveraged within our large-scale de novo design solutions to rapidly prioritize novel molecular architectures optimized against multiple objectives simultaneously.

Lastly, we highlight that these advances are underpinned by significant investments in the accuracy of our representations of the underlying physics. We will discuss the development of the first broadly applicable, highly accurate polarizable force field and the integration of sophisticated neural network interaction potentials to enable free energy calculations for systems previously considered intractable. Collectively, these integrated strategies provide a powerful, physics-driven approach for accelerating the design, optimization, and selection of promising development candidates.

Session 9: Industrial Applications & Development

O23 - The application of free energy methods at Bayer

Joseph Bluck

Bayer AG

Free energy calculations remain a cornerstone of the drug discovery toolkit, enabling the prediction of potency across diverse applicability domains. As we enter an era increasingly influenced by machine learning, these techniques are reshaping the landscape of drug discovery and decision making in compound design. In the context of free energy calculations, this is particularly prominent in the optimization of input structures, the extrapolation of results, or as a cost-effective alternative for extrapolating binding affinities. This talk will focus on our experiences at the interplay between traditional free energy methods and machine learning approaches for drug design.

O24 - From pilot to practice: Embedding FEP into DMTA cycles for prospective decision-making at UCB

Stephan Ehrlich

UCB Biopharma

The holistic de novo project is a digital transformation initiative at UCB to change the way new chemical entities (NCE) are discovered. With the ambition to arrive at better candidates faster, it builds on three key areas: Design, collaboration and predictions – leveraging the best available technologies in each.

Naturally, free energy methods (FEP) play a large role in the latter area, but while they have reached a level of maturity that enables their prospective use in drug discovery, sustained adoption within design–make–test–analyze (DMTA) cycles remains challenging.

The presentation will reflect on some of the challenges and successes we had on our journey of enabling FEP as a key decision-support capability across our NCE portfolio. We will highlight cases that include large, challenging targets, projects where the team struggled with imperfect predictions as well as a case where FEP helped to progress a backup series in the fraction of molecules synthesized on the lead series. To round it off, there is also an “off label” case where FEP was applied not purely as a potency predictor, and an example of our efforts to measure the bottom-line impact on our projects, recognizing the inherent difficulties associated with such evaluations.

025 - Addressing Electronic Complexity in FEP: Scaling QM/MM and Physics-Informed ML for Predictive Lead Optimization

David Pearlman, Jia Chen, David Reilley, Justin Provazza, Klaas Gunst

QSimulate

While classical force fields remain the bedrock of free energy perturbation (FEP), their reliance on empirically-fitted, fixed-charge models creates a significant "physics gap" in systems such as those involving formal charge changes, metals, anisotropic phenomena like pi-stacking and halogen interactions, and covalent binders. Historically, quantum mechanical (QM) methods were sidelined in drug discovery due to high computational overhead. However, recent algorithmic breakthroughs and GPU-optimized mixed-precision frameworks now enable QM/MM-FEP to reach industrial throughput, delivering predictive accuracy in hours rather than days.

This presentation details the evolution of the QM/MM workflow toward addressing complex electronic environments that are traditionally problematic for classical mechanics. We will discuss several key recent developments:

- Dynamic Solvent Treatment: A method for the dynamic reclassification of water molecules between QM and MM regions based on proximity to the ligand. This is critical for solvent-exposed active sites where h-bonding and water-bridge networks are sensitive to electronic treatment.*
- Physics-Augmented Machine Learning: An automated engine that derives bespoke, QM-quality force field parameters for non-canonical residues and macrocycles which are dropped directly into existing molecular mechanics workflows and provide significantly improved performance.*
- Covalent Lead Optimization: A rigorous QM/MM treatment suitable for covalent ligands series that share a warhead, enabling accurate binding affinity predictions where classical force fields struggle.*

026 - Physics-based and LQM-Driven Drug Discovery Applied to Neurodegeneration Drug Discovery

Kenneth Goossens, Steven Shave, Christoffer Lind, Jennifer Schreiber, Jialin Chen, Victor Sebastián Pérez, Peter Vangheluwe

SandboxAQ

The evolution of drug discovery is increasingly driven by advancements in AI, physics-based simulation, and multi-modal data integration. At SandboxAQ, we address fundamental challenges in the field by generating proprietary physics-based data to train Large Quantitative Models (LQMs).

To address the challenges in drug discovery, we have developed approaches that integrates AI protein structure prediction, AI binding affinity predictions, and physics-based methodologies to better understand protein flexibility and ligand affinity.

We will highlight and present the application of these methods to real project examples, showing the evaluation of novel drug discovery solutions for hit finding and hit-to-lead applied to the promising drug discovery targets for neurodegeneration.

In this work, we demonstrate the power of the LQM-driven framework with the application of our methods on a P-type ATPase implicated in Parkinson's Disease. This case study presented a significant challenge due to the protein's complexity, conformational flexibility, and lack of available structures. We report a comprehensive active-learning (AL) guided virtual screening workflow that integrates proprietary co-folding with a multimodal site identification strategy for hit finding.

By employing AL-docking and our proprietary free energy calculation method (AL-AQFEP), we efficiently prioritized compounds in a library of >5M compounds based on rigorous free-energy calculations. The approach resulted in the identification of allosteric pockets. This work provides a scalable framework for virtual screening on challenging membrane targets, demonstrating that combining active learning with physics-based methods can help successfully navigate chemical space for virtual screening on challenging targets.

Session 10: High-Throughput FEP & Scalable Workflows

O27 - Improved expanded ensemble methods for massively parallel virtual screening

Vincent A. Voelz

Temple University

In expanded ensemble (EE) approaches, the free energy of multiple thermostatic states are estimated using Monte Carlo transitions between ensembles with an adaptive bias. EE methods have advantages for massively parallel virtual screening because they enable alchemical free energy calculations to be performed in a single simulation replica, which is well-suited for distributed computing. Although EE has been underutilized in the field due to its sensitivity to the schedule of intermediates, our group has been working to develop improved EE approaches with better convergence properties and statistical accuracy. I will discuss these approaches, and how we have applied them to predict absolute and relative free energies at scale, in applications ranging from logP prediction, affinity maturation of computationally designed miniproteins, and peptidomimetic design.

O28 - Introducing NEMAT, a non-equilibrium method to calculate binding energies to membrane proteins

Ramon Crehuet

Institute for Advanced Chemistry of Catalonia (IQAC) - CSIC

Quantifying small-molecule binding to proteins is central to understanding biological mechanisms and to guiding rational drug design. Although alchemical free-energy methods now achieve high accuracy for soluble proteins, membrane proteins remain challenging due to the complex thermodynamic contributions of the lipid bilayer. Non-equilibrium free-energy perturbation (NEQ-FEP) strategies offer improved sampling efficiency and straightforward parallelization, yet no existing framework automates these calculations for membrane systems or enables systematic decomposition of membrane-partitioning and protein-specific contributions to binding.

Here we introduce NEMAT, an open-source workflow for automated non-equilibrium alchemical transformations in aqueous, membrane, and membrane-embedded protein environments. NEMAT standardizes system preparation, transition-path generation, and work-distribution analysis to ensure consistent sampling across environments. The pipeline provides explicit control over transition times, replica counts, and trajectory averaging, enabling robust overlap of non-equilibrium work distributions and reproducible free-energy estimates.

We demonstrate the capabilities of NEMAT by evaluating relative binding free energies for a series of P2Y₁ receptor ligands. NEQ-FEP calculations performed with NEMAT reproduce experimental affinity trends with accuracy comparable to established equilibrium methods while achieving improved computational efficiency. Importantly, the workflow's multi-environment design enables the decomposition of binding free energies into membrane-partitioning and protein-interaction components, providing mechanistic insight into ligand selectivity at lipid-exposed binding sites.

Together, these results show that NEQ-FEP can be applied reliably to membrane protein systems when supported by a consistent and automated workflow. By enabling mechanistically interpretable affinity predictions and facilitating routine application of non-equilibrium alchemical approaches, NEMAT advances the practical use of free-energy calculations in the study and design of ligands targeting membrane-embedded proteins.

O29 - Scaling LaDyBUGS for Drug Discovery: Methodological Enhancements and Expanded Protein–Ligand Benchmarks

Jonah Vilseck

Indiana University School of Medicine

Accurate and efficient alchemical free energy calculations are essential for modern drug discovery, where large ligand libraries must be screened quickly and reliably. To address this challenge, we have developed λ -Dynamics with Bias Updated Gibbs Sampling (LaDyBUGS), a method that merges the flexibility of λ -dynamics with the rapid λ -state hopping capabilities of Gibbs sampling and continuously updated biases. This combination enables LaDyBUGS to sample tens to hundreds of discrete λ states efficiently to evaluate many ligand modifications simultaneously within a single simulation. Previous benchmark studies have demonstrated that LaDyBUGS can be 35–100 times more efficient than classic Thermodynamic Integration (TI) or MBAR while maintaining high accuracy, typically within 0.3–0.5 kcal/mol of TI calculations and near or below 1.0 kcal/mol relative to experiment. Building on these results, we have been working to further expand and refine the LaDyBUGS method. Ongoing work includes extending benchmarks to a broader range of protein–ligand systems, exploring alternative λ schedules to optimize alchemical transitions, and developing a newly parallelized implementation of LaDyBUGS that improves scalability on distributed GPU systems. This presentation will highlight these advances, discuss their impact on accuracy, throughput, and applicability, and review challenges encountered along the way. Overall, this work demonstrates that LaDyBUGS is a powerful and increasingly versatile tool for accelerating computational design and free energy prediction.

O30 - Chimeric Atom Matching and Sampling Sufficiency as Determinants of RBFE Accuracy

Chris Neale, Hyesu Jang, Arghya Chakravorty, Victor H. Chavez, Caitlin Bannan, Gaetano Calabro

OpenEye, Cadence Molecular Sciences

Chimeric atom matching is a critical and often underappreciated driver of RBFE performance. Using case studies, we show how atom-mapping choices affect RBFE accuracy through the deliberate selection of chemically appropriate (rather than maximal) common substructures informed by 3D information, the avoidance or mitigation of valence-term strain, and the preservation of handedness and chirality. These examples indicate that atom-mapping choices can be a primary driver of practical accuracy, in some cases outweighing gains achieved through increased sampling. We further show that extensive sampling is not always required to achieve actionable RBFE predictions during iterative design on congeneric series. By trading modest reductions in precision for substantial gains in throughput, a fixed computational budget can be used to evaluate a larger number of ideas, accelerating design cycles and improving overall discovery efficiency.

POSTER PRESENTATIONS

Poster Session 1

P1 - NEMAT: An Automated Non-Equilibrium Free-Energy Framework for Predicting Ligand Affinity in Membrane Proteins

Albert Ortega-Bartolomé, Ramon Crehuet

Institut de Química avançada de Catalunya (CSIC)

Quantifying the strength of small-molecule binding to proteins is essential for understanding biological function and for advancing drug discovery. Computational free-energy methods aim to predict these interactions accurately, yet membrane proteins remain particularly difficult due to the added thermodynamic effects of the lipid bilayer. Alchemical free-energy calculations provide state-of-the-art accuracy by transforming one ligand into another, and recent non-equilibrium approaches (NEQ-FEP) offer improved efficiency and parallelization. However, no existing workflow automates NEQ-FEP for membrane proteins and enables decomposition of the observed binding free energy into membrane-partitioning and protein-specific components. NEMAT is an open-source pipeline that performs automated non-equilibrium alchemical transformations in water, membranes, and membrane-embedded protein environments.

Here we show that NEMAT reproduces experimental binding-energy trends for P2Y1 ligands with accuracy comparable to established equilibrium methods. These findings demonstrate that NEQ-FEP can be applied reliably to membrane systems when supported by a consistent workflow. In contrast to previous approaches, NEMAT provides systematic control over transition length, sampling, and replica averaging, enabling stable work-distribution overlap and reproducible free-energy estimates. In a broader context, NEMAT offers a practical route to mechanistically interpretable affinity predictions for membrane-embedded targets. Its ability to dissect membrane and protein contributions advances our understanding of ligand selectivity at lipid-facing pockets and supports the routine application of non-equilibrium free-energy methods in drug discovery.

P2 - Training a force field for proteins and small molecules from scratch

Alexandre Blanco-González, Thea K. Schulze, Evianne Rovers, Joe G. Greener

MRC - Laboratory of Molecular Biology

Force fields used in Molecular Dynamics have historically been developed manually, limiting their transferability and making the systematic exploration of functional forms challenging. In this work we develop a Graph Neural Network that assigns all force field parameters for diverse molecules from scratch, using continuous atom typing.

The model, called Garnet, was trained on quantum, condensed phase, and protein NMR data; without the use of existing force field parameters. The resulting force field shows comparable performance to existing parameterisations on small molecules, folded proteins, disordered proteins and protein complexes. It shows similar results to popular approaches for binding free energy predictions across a range of targets. The assesment of different functional forms suggests that the double exponential potential is a flexible and accurate alternative to the Lennard-Jones potential.

These results show that automated, reproducible force field development pipelines will be important if classical force fields are to benefit from the machine learning age.

P3 - Open Free Energy: An Ecosystem for Open Source Alchemy

Alyssa Travitz, Irfan Alibay, Hannah M. Baumann, David L. Dotson, James RB Eastwood, Mike M. Henry, Josh Horton, Ian M. Kenney, Iván Pulido, Benjamin Ries, David Mobley

Open Free Energy, OMSF

Alchemical free energy calculations are integral to state-of-the-art drug discovery pipelines, yet the need for domain expertise and reliance on closed-source software remain major barriers to their widespread use. The Open Free Energy (OpenFE) project is a pre-competitive collaboration between academic and industry partners to develop an ecosystem of extensible open-source tools for accessible and reproducible free energy calculations. OpenFE's ecosystem includes tools for network planning, atom mapping, and defining protocols for relative and absolute alchemical transformations, simulation execution, and data analysis. We will present an overview of the software libraries within the OpenFE ecosystem and highlight recently added capabilities, as well as demonstrate how these tools can be used out-of-the-box, adapted, or extended to create end-to-end workflows enabling robust free energy calculations at scale.

P4 - Anchor-Based Alchemical Transformations using Neural Network Potentials

Anna Katharina Picha, Stefan Boresch

University of Vienna

Free energy simulations (FES) often rely on alchemical transformations. Many such methods are well established in the context of traditional force fields. In recent years, neural network potentials (NNPs) have undergone rapid development, and their potentially higher accuracy makes them attractive for use in FES. In previous work, we demonstrated how molecules can be gradually decoupled in systems fully described by NNPs by manipulating the neighbor list and introducing an offset to interatomic distances. This methodology was successfully applied to compute absolute solvation free energies. Here we generalize this approach to alchemical transformations between, in principle, arbitrary pairs of molecules. As in our earlier work for decoupling molecules from their environment, we manipulate the neighbor list to control interactions during the alchemical transformation. Therefore, as in the case of absolute solvation free energies, the method is, in theory, agnostic to the underlying NNP architecture. In addition, our approach does not necessarily require a high degree of chemical similarity between the molecules being transformed. In the dual coordinate dual topology paradigm used, the two solutes do not share a common core or any atoms; instead, both solutes are fully represented throughout the simulation. Conceptually, the alchemical transformation corresponds to continuously converting the interactions of one solute with the surrounding solvent into those of another solute with the same solvent environment. To achieve this, it is essential that both solutes mostly interact with the same set of solvent molecules during the transformation. We enforce this by spatially overlapping the solutes, disabling direct interactions between them, and applying positional restraints via additional force-field terms to a fixed position, which we refer to as anchor. We validate the correctness of the methodology by demonstrating thermodynamic cycle closure for several solute pairs and report relative free energies computed using the MACE-OFF23(S) neural network potential.

P5 - Modelling Macrocyclic Peptide PCSK9 Inhibitors with Alchemical Free Energy Calculations

Audrius Kalpokas, Mark Mackey, Julien Michel

University of Edinburgh

Free energy perturbation (FEP) methodologies are widely used in drug discovery campaigns to optimise the binding affinity of small molecule ligands. Recently, macrocyclic peptides have emerged as an increasingly attractive modality alongside these traditional therapeutics. They possess the unique ability to bind to large, flat, and featureless binding sites inaccessible to small molecules, whilst maintaining the oral bioavailability often lacking in antibodies. However, modelling macrocycles with FEP methodologies presents significant challenges. These campaigns frequently involve complex scaffold changes that extend beyond traditional R-group modifications, and often utilize both canonical and non-canonical amino acids, thus necessitating specialized parametrization protocols to handle such diverse chemical space.

In this work, we address these challenges by benchmarking against the seminal Tucker et al. study [1], which details the development of novel macrocyclic peptide inhibitors targeting PCSK9. We present open-source, state-of-the-art relative binding free energy (RBFE) protocols capable of robustly handling the complex scaffold modifications found in this dataset, such as ring-breaking, linker reconnection, and cross-linker addition for example. Furthermore, we evaluate and refine automated parametrization protocols to accurately model these chemically diverse inhibitors. Taken together, this work broadens the domain of applicability of FEP methodologies in drug discovery.

References

[1] - Tucker et al. A Series of Novel, Highly Potent, and Orally Bioavailable Next-Generation Tricyclic Peptide PCSK9 Inhibitors. *J. Med. Chem.* 2021, 64 (22), 16770–16800. <https://doi.org/10.1021/acs.jmedchem.1c01599>.

P6 - Exploring the relevance of structure-based machine learning in binding affinity predictions

Benjamin Kaminow, Alexander Matthew Payne, Hugo MacDermott-Opeskin, Sukrit Singh, John Chodera

Memorial Sloan Kettering Cancer Center

Structure-based drug discovery is a widely used paradigm for the rational design of novel small molecule therapeutics. However, this utilization of structural information has not translated to machine learning, where ligand-only models are still the industry standard. Here, we rigorously compare structure-based ML models for predicting binding affinity with 1D and 2D baselines, considering different model architectures, training methods, and prediction tasks. To assess different methods of incorporating physicality into the model prediction, we decompose the prediction task into three steps, allowing us to fine-tune the way in which structural information is utilized by the model. Additionally, we evaluate both $E(3)$ -invariant and $E(3)$ -equivariant architectures to determine the importance of these inductive biases.

By first optimizing several aspects of model construction using a general purpose dataset, we are able to improve the performance and data efficiency of 3D models; when subsequently trained and evaluated on the COVID Moonshot drug discovery dataset, these tuned models perform on par with industry standard ligand-only models. Through this process, we find that incorporating physicality into model predictions improves performance across all tested architectures. We also find that including equivariance does not improve model performance relative to the invariant architecture, however equivariance does appear to improve the model's ability to generalize from a pretraining dataset.

P7 - Molecular glues in drug design: enhancing protein-protein interaction stability for innovative cancer therapies

Díaz-Canals, Blanca; Juárez-Jiménez, Jordi

Universitat de Barcelona

Molecular glues (MGs) could bring about a paradigmatic shift in the fields of medicinal chemistry and drug design. However, to fully harness the therapeutic potential of MGs is pivotal to understand their role in the stability of the protein-protein interface. In line with our group's commitment to advancing computational tools for complex therapeutic targets, our efforts are directed towards uncovering the physicochemical principles governing these interactions. We aim to integrate these principles into new computational tools to identify small molecules that excel in stabilizing ternary complexes.

Our test case is the E3 ligase Cereblon (CRBN), which is an essential component of the current efforts to generalize the Targeted Protein Degradation approach. During my thesis, emphasis is placed in discovering small molecules that are able to induce the degradation of proteins overexpressed in cancer providing a promising avenue for innovative anticancer strategies. This project builds on findings from a recent paper (1) showing that lenalidomide stabilizes the proteinprotein complex between cereblon and CK1a by increasing the strength of hydrogen bonds between the proteins.

References:

[1] Miñarro-Lleonar, M.; Bertran-Mostazo, A.; Duro, J.; Barril, X.; Juárez-Jiménez, J. J. *Med. Chem.* 2023, 66, 6037–6046.

P8 - Computational Alanine Scanning of Conotoxin Peptides as a Route to Antitoxin Design

Charlie Holdship, Prof. Jeremy Frey and Prof. Jonathan Essex

University of Southampton

Conotoxin peptides are a class of neurotoxin derived from marine cone snail venom that target human nicotinic acetylcholine receptors (nAChRs) and cause paralysis upon envenomation. Currently, no known antitoxins exist, and while conotoxin solution-phase conformations and nAChR binding sites are known, until recently, their precise binding modes were not. Recently, the structures of three different conotoxins bound to muscle type nAChR (*T. californica*) have been solved through cryo-EM, providing new insight into the toxin/receptor interactions, at each of the distinct nAChR acetylcholine binding sites.[1]

To further explore the role of these interactions in determining affinity and specificity, we use free energy methods to perform a computational alanine scan of these peptides to identify which residues are most important for binding. We investigate the difference in the importance of residues for binding at each of the two nAChR binding sites. We also compare the differences between traditional equilibrium free energy techniques (free energy perturbation – FEP) and non-equilibrium methods (non-equilibrium switching – NES) to see which is better suited for this type of system.

Through comparison with experiment, we identify which residues are essential for binding to the nAChR and further our understanding of these toxins' mechanisms of action and binding site specificity.[2] Knowledge of these residues is crucial for the de novo design of conotoxin antitoxins, work which is currently in progress.

References:

[1] Capper, M., Shepperson, O., Holdship, C., Melling, O., Wade, N. Malone, M., Arnott, K., Windeln, L., Turner, S., Whitmore, C., Morcom, T., Connah, J., Timperley, C., Frey, J., Green, C., Essex, J., Köhnke, J., & Jamieson, A. *bioRxiv* 2026.01.15.699711; doi: <https://doi.org/10.64898/2026.01.15.699711>

[2] Ning, J., Li, R., Ren, J., Zhangsun, D., Zhu, X., Wu, Y., & Luo, S. (2018). *Mar Drugs.*, 16(12), 507. doi: <https://doi.org/10.3390/md16120507>

P9 - Enhancing Relative Binding Free Energy Calculation with Grand Canonical Monte Carlo, Water Swap Monte Carlo, and Replica Exchange Solute Tempering

Chenggong Hui; Bert de Groot

Max planck institute for multidisciplinary sciences

Alchemical free energy calculations, particularly free energy perturbation (FEP), have become routine in drug design. When molecular dynamics (MD) is employed as the sampling method in FEP, accurately sampling water molecules deeply buried in protein binding pockets remains challenging, despite their critical role in protein-ligand interactions. We implemented three enhanced sampling methods, Grand Canonical Monte Carlo (GCMC), water swap Monte Carlo (water MC), and replica exchange solute tempering (REST2). GCMC or water MC accelerate the hydration state equilibration in buried pockets and REST2 accelerates the sampling of ligand conformations. Combining REST2 with GCMC or water MC enables accurate calculation of ligand relative binding free energies. In the benchmark on the water set, our package achieved an RMSE of 0.94 kcal/mol, compared to 0.86 kcal/mol from FEP+ and 1.60 kcal/mol from OpenFE. This development provides the scientific community with an open-source sampling engine for robust and accurate FEP calculations.

P10 - Decoding the Rules of Phase Separation through alchemical transformations on model peptides

David De Sancho, Xabier López & Lydia Armentia

University of the Basque Country

Short peptides have been instrumental as models for more complex systems. Studying short peptides, we have gained understanding of fundamental folding events like alpha helix nucleation or hairpin folding, and in the determination of the microscopic origin of internal friction. Short peptides are also challenging tests for the calibration of modern force fields, and hence have been extensively used in recent optimization efforts. In our laboratory, we have recently investigated these peptides as simplified model systems to study intrinsically disordered proteins (IDPs). Examples of our recent work include probing how metal ions influence their conformational dynamics and exploring the potential of cyclic peptides to serve as molecular binders.

In my talk I will focus on our recent work on phase separation. Using peptide models, we test the hypothesis that phase separation is an emergent property determined by composition, even in the absence of a polypeptide chain, multivalency and patterning effects. For this, we use atomistic molecular dynamics (MD) simulations of saturated solutions of individual amino acids and mixtures thereof in stoichiometries comparable to those of phase-separating low complexity domains. Additionally, we have used alchemical transformations using a established non-equilibrium formalism to disentangle the hierarchy of interaction strength between the two most dominant types of aromatic and positively charged residues in condensates. Our results are broadly consistent with trends observed in experiments and in atomistic simulations of full-length IDPs and reconcile findings from decades of work in physical chemistry and protein biophysics.

P11 - Efficient and fully automated large scale execution of alchemical campaigns with alchemiscale

David L. Dotson, Ian M. Kenney, Jan Domanski, Hugo MacDermott-Opeskin, Ivan Pulido, Mike Henry, Irfan Alibay, Jenke Scheen, Hannah Baumann, Josh Horton, Alyssa Travitz, Jeff Wagner, John D. Chodera

Datryllic LLC

Commercial solutions for performing scalable alchemical free energy calculations (FECs) often limit their usefulness with prohibitive licensing that scales with utilization. For drug discovery programs, this adds substantially to the total cost of computational chemistry, and for resource-constrained organizations can seriously limit the degree to which FECs can be used to explore chemical space under tight deadlines.

alchemiscale offers a new approach. Fully open-source, and built on the Open Free Energy software ecosystem, alchemiscale is a distributed execution system that can scale FECs efficiently and autonomously across a variety of compute resources (in-house clusters, spot on the cloud). With its permissive MIT license, alchemiscale can be freely deployed and extended to suit and scale to available infrastructure at no additional cost. In use by OMSF and several biotechs for production work since 2023, alchemiscale has since matured into an industry-grade execution system.

We demonstrate alchemiscale's ease of use end-to-end: deployment to cloud resources, preparation and submission of alchemical networks, automated compute allocation and scaling, introspection of compute progress, and analysis of FEC results as they become available.

P12 - An Explainable and Uncertainty-Aware Bayesian Graph Neural Network for Predicting the effect of Protein mutations

David Sjöberg, Lucien Koenekoop, Johan Åqvist, Hugo Gutierrez de Teran

Uppsala University

Accurate predictions of the shifts in thermal stability caused by amino acid substitutions are key to understand the molecular mechanisms of disease-related variants, or to guide protein engineering. In recent years, machine learning methods (ML) have matured and show very competitive accuracy for prediction of thermal stability (expressed as $\Delta\Delta G$). However, these family of methods are relatively opaque in the sense that they provide little information about the confidence of the predictions and lack structural interpretation.

To tackle these limitations, we developed a Siamese Bayesian Graph Neural Network (SB-GNN), which operates on a collection of descriptors combining information from chemical topology with physics-based potential energy for both wild-type and mutant graphs.

The network was trained on a curated subset of the MegaScale cDNA proteolysis dataset ($n = 226\,000$) (Tsuboyama et al, 2023). It was subsequently evaluated on a subset of the dataset ensuring low homology with respect to the training data, as well as on common benchmarking sets, to ensure the robustness of the network. The resulting SB-GNN model can learn the effects in protein stability not only for single, but also for double mutants. Additionally, we incorporated the post-hoc method GNNExplainer (Ying et al, 2019) to identify the driving structural and chemical elements most influential for each prediction.

The resulting SB-GNN model achieves very high predictive performance ($MAE = 0.7$ kcal/mol, Pearson $r = 0.7$) on our internal test set ($n=49\,000$), comparable to current deep learning methods, with the additional advantage of allowing user interpretation and uncertainty estimation. The uncertainties allow for user-based filtering and can help detect cases where physics-based free energy perturbation (FEP) calculations show large errors, enabling combined use of both approaches for strong predictive performance.

P13 - Nonequilibrium Chimeric Switching (NEX) for parallelizable Relative Binding Free Energy Calculations

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SandboxAQ

The accurate and rapid prediction of protein-ligand binding affinities remains a critical bottleneck in computational drug discovery. While alchemical free energy methods offer high accuracy, their computational expense often limits their applicability in fast-paced design cycles. Here, we introduce Nonequilibrium Chimeric Switching (NEX), a novel method that combines rapid nonequilibrium trajectories with a dual-topology Alchemical Transfer Method (ATM) framework to efficiently calculate relative binding free energies. Our results show that NEX is highly performant, matching the accuracy of traditional ATM simulations with a framework optimized for modern distributed computing. We can readily leverage a massively parallel architecture, significantly increasing computational throughput, and reducing the wall-clock time required for large-scale screenings. A key finding is that the method's robustness and convergence are significantly improved by employing an intermediate chimeric state as a central hub for the nonequilibrium switches, rather than performing a direct transformation between the physical nonalchemical endstates.

By providing a framework that is fast, accurate, and broadly applicable to diverse chemical matter, we anticipate that NEX will be a valuable methodology for accelerating molecular design efforts in both preclinical hit-to-lead and lead optimization campaigns.

P14 - Exploration of alternative λ -networks in LaDyBUGS (λ -Dynamics with Bias Updated Gibbs Sampling) to optimize sampling efficiency

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The accurate prediction of protein-ligand binding free energies with alchemical free energy (AFE) methods has become a powerful approach for computationally investigating lead compound analogs prior to experimental assessment, thereby reducing costs and accelerating structure-based drug design. The Vilseck lab has recently developed LaDyBUGS (λ -Dynamics with Bias Updated Gibbs Sampling), a new AFE method that has shown measurably greater throughput compared to conventional methods. Previous benchmark studies have shown LaDyBUGS to be 18 to 200 times more efficient and as equally accurate as conventional AFE methods for protein-small molecule systems. In these studies, LaDyBUGS used a “strongly connected” network of discrete λ states to facilitate alchemical transitions between 2 to 7 different molecules within a single simulation. This eliminated the need to perform redundant cycle closure calculations, as typically done for pairwise-based methods which require careful attention to the selection of pairs and edges connecting molecular analogs. Although LaDyBUGS performs well as implemented in CHARMM, further optimization of its λ -network could yield enhancements in computational efficiency and accuracy. For example, practical limitations currently prevent the sampling of more than 10 ligands simultaneously in a LaDyBUGS calculation. This means extra time must be spent on organizing ligands before running simulations. Thus, this work aims to explore alternative λ -networks and associated discrete λ states with LaDyBUGS to optimize its sampling efficiency. Ideally, these alternative λ -networks will reveal if LaDyBUGS can accurately and simultaneously sample larger perturbation sets (≥ 10) within a single simulation. This work has begun by calculating ligand binding affinities to Major urinary protein 1 (MUP1), a popular benchmark system, and will be extended to additional literature benchmark systems. Several alternative λ -network mappings will be performed and compared to the native “strongly connected” λ -network to gauge improvements. These and future benchmarking results will be discussed in this presentation.

P15 - Scalable Multistate Free Energy Calculations with automated Network Design

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Replica-exchange enveloping distribution sampling (RE-EDS) is an efficient method for estimating free-energy differences among multiple thermodynamic states simultaneously, yet its application to large ligand libraries of arbitrary complexity requires the development of an automated, end-to-end framework that streamlines the setup and execution of relative binding free energy calculations specifically for multistate methods. Large datasets are first partitioned into alchemically coherent sub-groups with a clustering algorithm using simple similarity metrics while maintaining full connectivity of the molecular graph. The high similarity within these clusters ensures improved sampling efficiency of the relevant phase space in the multistate setup compared to pairwise methods. We further improve end-state transitions within each cluster by introducing soft-core potentials to the Hamiltonians of the biased replicas, facilitating transformations involving many atom insertions or deletions and complementing the standard smoothness bias. The robustness of this pipeline is rigorously validated against the Protein-Ligand Benchmark dataset for free energy calculations, marking the first large-scale assessment of the OpenMM-based implementation of RE-EDS for relative binding free energies. The results demonstrate that combining automated network design with soft-core enhanced sampling allows RE-EDS to reproduce established binding affinities with high accuracy while incurring a fraction of the computational cost of traditional pairwise approaches.

P16 - Targeting Undruggable Proteins Using ISOKANN framework

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A proteome is considered as druggable if the drug molecule finds a well-defined hydrophobic pocket in a protein and stays there to block its function. A target is considered undruggable if it lacks this pocket.

Although cryo-EM has been a game-changer for the undruggable targets, it leaves us an incomplete picture of how a protein would behave in a living system. Therefore, we use in silico methods to convert the frozen protein into a part of a living and moving organism.

One can model and explain the chemical reactions or conformational transitions with a free energy landscape, for which we need to have a well-defined reaction coordinate. Proteins often spend most of the reaction time in metastable states, whereas the most important parts of the reactions are called the rare events. With an effective reaction coordinate, we can identify the metastabilities and the rare events, and thus show the progress of the reaction along the coordinate.

If we know the metastabilities of the biomolecule, we can model the reaction using the committor function from Transition Path Theory. However, identifying the rare events for undruggable targets still remains cryptic due to the unavailability of apriori information about the protein.

In this project, we utilized a neural-network augmented algorithm – ISOKANN – to calculate a reaction coordinate with no apriori information about the target. ISOKANN, standing for Invariant Subspaces of the Koopman Operator learned by Neural Network, calculates a membership-valued reaction coordinate, which carries information from the dominant eigenfunctions of the transfer operator, thereby the slowest process in the reaction.

Then we isolated specific modes of the protein-ligand complex by further clustering the state-space using a topology-based algorithm, MoKiTo (Molecular Kinetics via Topology). By using such a clustering method, we could also visualize multiple enthalpic and entropic transition pathways through the free energy surface.

P17 - Fast training of Open Force Field valence parameters for free energy calculations

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Molecular mechanics force fields enable the extensive sampling required for accurate alchemical free energy calculations. While transferable force fields balance performance across diverse molecules, molecule-specific "bespoke" parameters can substantially improve accuracy for individual systems. The Open Force Field BespokeFit tool [1] automates bespoke torsion fit-ting to quantum mechanical data, but the approach is slow, requiring fragmentation and torsion scans for each rotatable bond, and optimises only proper torsions while leaving other valence parameters (bonds, angles, and improper torsions) untouched.

We present a significantly faster and more comprehensive approach to bespoke parameterisation. Our method generates training data using high-temperature molecular dynamics with well-tempered metadynamics and iteratively fits all valence parameters to energies and forces from a machine learning potential (MLP). By leveraging fast MLP evaluations and eliminating the need for systematic torsion scans, parameterisation completes in a few minutes for a 50-atom molecule on a single GPU. Despite the absence of explicit torsion scans, the resulting force fields reproduce accurate torsional profiles.

We demonstrate over 50 % reductions in torsional energy profile errors compared to Open Force Field's 'Sage', a leading open-source force field [2]. We highlight a protocol for the simultaneous fitting of congeneric series and discuss the performance of the resulting force fields in free energy calculations. Our open-source implementation (<https://github.com/cole-group/presto>) enables researchers to rapidly generate reliable, molecule-specific force fields that maintain the computational speed of traditional molecular mechanics force fields while improving accuracy of conformational sampling.

[1] Horton, J. T. et al. (2022). *J. Chem. Inf. Model.* 62(22), 5622–5633.

[2] Boothroyd, S. et al. (2023). *J. Chem. Theory Comput.* 19(11), 3251–3275.

P18 - Fast Free Energy Prediction with FE-NES

Gaetano Calabro, Chris Neale

OpenEye Cadence Molecular Sciences

A variety of free-energy calculation methods including FEP, TI, and λ -dynamics employ atomistic molecular dynamics or Monte Carlo simulations to estimate relative ligand binding free energies. These approaches are now widely integrated into pharmaceutical lead-optimization workflows to quantitatively assess the impact of chemical modifications on binding affinity. Free-energy nonequilibrium switching (FE-NES) has more recently emerged as an alternative methodology that estimates binding affinities by computing the thermodynamic work required to transition between equilibrium states via rapid alchemical transformations. The accuracy and efficiency of FE-NES calculations depend critically on several tunable parameters to efficiently navigate the alchemical space, particularly the switching duration and the number of sampled equilibrium configurations. In this study, we evaluate how these parameters influence predictive performance within the Orion® cloud computing platform, with an emphasis on scenarios that require high-throughput assessment of large sets of in-silico design hypotheses

P19 - Scaling Absolute Binding Free Energy Calculations for Virtual Screening

Haolin Du, Marley Samways, Richard Taylor, Julien Michel

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Alchemical absolute binding free energy (ABFE) calculations are accurate yet computationally expensive tools for predicting ligand–protein binding affinities. While ABFE provides high accuracy,¹ its low throughput limits its application in large-scale virtual screening. In contrast, traditional virtual screening methods using docking or machine learning offer high throughput but often suffer from limited predictive power.

To bridge this gap, we present a FastABFE protocol based on the A3FE software package^{2,3} that uses ca. $\times 10$ - $\times 100$ fold shorter simulation time than established ABFE protocols. We show that although the computed binding free energies deviate systematically from experimental data, reasonable qualitative rankings can be retained.

FastABFE was benchmarked on a dataset of congeneric series of 216 protein-ligand complexes for which extensive benchmarks were previously reported by Bayer.⁴ We show that FastABFE rankings approach the rankings obtained by relative binding free energy methods such as PMX.

We have also evaluated the performance of FastABFE in the context of virtual screening, by using a dataset of 1162 PDE10A ligands published by Roche,⁵ that we supplemented with 9574 DUD-E generated decoys. We discuss considerations such as relationship between accuracy of predicted poses and reliability of affinity rankings.

P20 - In silico-driven protocol for hit-to-lead optimization: a case study on PDE9A inhibitors

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Hit-to-lead (H2L) is a critical early stage of small-molecule drug discovery. Identifying high-quality leads at this stage markedly improves overall efficiency and success rates. Conventional H2L relies on experimental, iterative cycles of synthesis and biological evaluation; practical limits on synthesis restrict the explored chemical space, making multi-objective optimization difficult. Consequently, medicinal-chemistry decisions become a trade-off problem: improving potency often compromises solubility, metabolic stability, or other ADMET properties. In silico H2L approaches have therefore attracted increasing interest. They generate large virtual libraries and prioritize compounds using predicted binding affinities and ADMET properties, enabling a more exhaustive exploration of chemical space. Here we present an in silico-driven protocol that integrates (i) rule-based systematic molecular generation, (ii) high-accuracy relative binding free energy (RBFE) calculations via non-equilibrium switching (NES), and (iii) machine-learning ADME predictions. As a case study we applied the workflow to a Bayer hit-to-lead campaign that ultimately delivered BAY-7081, a phosphodiesterase 9A (PDE9A) inhibitor with high potency, good aqueous solubility, and a favorable pharmacokinetic profile. First, scaffolds that offered a balanced starting point for potency and ADME were selected. Next, substituents that engage a hydrophobic pocket were introduced to improve potency. Finally, solvent-exposed groups were modified to improve ADME characteristics. The literature-reported lead ranked among the top candidates, showing that the workflow can recapitulate a successful H2L outcome. Importantly, compounds that docking alone missed were appropriately evaluated by NES-based RBFE calculations; their selection was then refined with predicted ADME properties. Our results demonstrate that RBFE-based virtual screening, combined with systematic R-group enumeration and ML-based property prediction, enables simultaneous improvement of potency and ADME, reduces synthetic effort, and accelerates decision-making in prospective H2L campaigns. The workflow is transferable to other targets, offering a practical platform for early-stage multi-parameter drug design.

P21 - Rational iterative structure-based drug design in Orion

Hyesu Jang, Kalistyn Lemke, Arghya Chakravorty, Paul Hawkins, Chris Neale

OpenEye, Cadence Molecular Sciences

We present a highly automated, hypothesis-driven in silico workflow in the Orion® cloud platform that enables medicinal chemists to make rapid and informed design decisions during lead optimization. The workflow integrates three components into a continuous iterative cycle: (1) generation of ligand-modification hypotheses, (2) fragment replacement-based molecule design, and (3) quantitative assessment of free energies for the proposed analogs. Hypotheses are first generated using GamePlan, which analyzes binding-site features by sampling various probe water orientations to generate ligand modification hypotheses. These hypotheses are translated into concrete design ideas through automated fragment replacement-based molecule generation in BROOD. The resulting analogs are then evaluated using fast, scalable free energy calculation method, FE-NES in Orion to assess the viability of each proposed design. By coupling automated insight generation with predictive scoring, this workflow provides medicinal chemists with actionable guidance on what to synthesize and what to avoid, reducing design cycles and focusing experimental resources on the most promising ideas. The result is a practical, efficient decision-support system for rational, iterative structure-based drug design.

P22 - Exploring the sensitivity of relative binding free energy calculations to force field choices

Irfan Alibay, Josh Horton, Hannah M. Baumann, James RB Eastwood, Mike M. Henry, David Mobley, Alyssa Travitz

Open Free Energy, The Open Molecular Software Foundation

As demonstrated by Fleck et al. (JCTC, 2021), the presence of redundant valence terms introduced by dummy atoms in certain alchemical topology schemes can lead to incorrect dynamics and inaccurate free energies. To address this issue, several correction schemes have been introduced; ranging from naive scaling of coupling terms to implementing automated rules for identifying and removing redundant terms following the recommended best practices.

In this contribution, we investigate the impact of these different correction schemes within a hybrid topology framework as implemented in the Open Free Energy toolkit. We assess their performance using a hierarchy of diagnostics workflows and metrics, including geometrical disagreement of minimised structures, thermodynamic cycle closure analyses and binding free energy benchmarks.

P23 - Free Energy Calculations at Johnson & Johnson: Small Molecules, Modalities, Off-targets.

David Hahn, Laura Perez Benito, Vytas Gapsys

J&J

In drug discovery, the assessment of binding affinities through free-energy calculations is becoming essential for selecting the most promising compounds for synthesis. We present the integration of FE into the in-silico compound optimization workflows at Johnson & Johnson, enabling us to evaluate calculated property profiles of compounds prior to synthesis in a "predict first" approach. This presentation addresses the challenges encountered during this integration, particularly in transitioning from small-scale retrospective literature examples to large-scale prospective calculations. Likewise, we will give insights into the transition from relative binding free energy calculations of small-molecules to larger molecules like peptides or proximity inducing therapeutics like PROTACs. Beyond the calculation of binding affinities, we present how free energy methods have been proven to be a versatile tool in the assessment of hypothesized binding sites or off-target activities.

P24 - Exploring the chemical features driving affinity in a K2P channel activator

Edward Francisco Mendez Otalvaro, Bert L. de Groot

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Modulating ion permeation in potassium (K^+) channels, particularly at the selectivity filter (the narrowest region of the pore), is of significant pharmaceutical interest. Experiments have shown that the two-pore domain K^+ channel TREK1 undergoes C-type inactivation (i.e., distortions in the filter structure that hinder ion permeation). In addition, in previous simulations we have shown that a carbonyl flip, which disrupts K^+ coordination in the filter, can also impede ion conduction.

Both mechanisms are prevented by a small-molecule activator bound adjacent to the filter, highlighting opportunities for the design of improved activators targeting these gating processes. We hypothesize that polar or charged groups in the upper region of the activator enhance binding and likely reduce C-type inactivation, whereas larger, more hydrophobic groups in the lower region could further increase binding affinity and suppress the carbonyl flip.

By probing representative chemical substitutions using calibrated non-equilibrium free-energy calculations, we test these hypotheses and identify a structure–affinity relationship, that suggests the most appropriate physicochemical features for improving modulator binding to this K^+ channel.

P25 - Structural characterization of cannabidiol and cannabigerol orthosteric and allosteric binding to the adenosine A3 receptor

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Computational Biochemistry Unit (CINN/CSIC)

Cannabidiol (CBD) and cannabigerol (CBG) are phytocannabinoids found in the plant *Cannabis sativa* that do not exhibit psychotropic effects and have potential therapeutic applications. In particular, CBD is already approved in the U.S. in the form of EpidiolexTM for treating seizures associated to several rare diseases. While the mechanism of action of cannabinoids have been associated to cannabinoid receptors, different targets have been identified, such as serotonin 5-HT_{1A} or μ -opioid receptor. Recent studies have shown that CBD is also an allosteric modulator of adenosine receptors. Here, we investigated the binding mode of CBD and CBG on adenosine A₃ receptor (A₃AR). Given the known allosteric effect of CBD on its targets, potential allosteric sites in A₃AR were explored using FTMap cavity detection, which identified allosteric pockets previously reported for different G protein-coupled receptor (GPCR) allosteric modulators. To cross-validate these sites, 1 μ s molecular dynamics (MD) simulations of inactive and active apo-G α -bound A₃AR were performed, and allosteric communication was evaluated based on mutual information network analyses, which were consistent with the identified allosteric cavities. Subsequent docking and MD simulations of CBD and CBG on these sites support binding to either the orthosteric site or an allosteric site that overlaps with a conserved cholesterol binding site in GPCRs and a cannabinoid CB₁ receptor allosteric binding site. Finally, *in silico* mutagenesis was performed using free energy perturbation with QresFEP to rationalize experimentally observed efficacy changes of the reference A₃AR agonist IB-MECA in presence of CBD or CBG. These results provide a structural framework for cannabinoid orthosteric and allosteric binding to A₃AR and suggest potential sites for pharmacological modulation.

P26 - Scaling of non-bonded inter-molecular interactions improves convergence in the geometric route for protein-protein binding free energy calculations

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Proteins are among the most studied entities for use as novel drugs due to their ability to bind to targets considered undruggable by small molecules. However, the use of computational methods to predict binding free energies of protein-protein systems faces significant challenges.

Due to the large number of atoms, the alchemical route is typically not feasible for protein-protein systems. Instead, most current approaches utilize the geometric route in which the proteins are spatially separated, recovering the free energy from the separation process. In addition, restraints on the orientation, rotation, and conformation are used to limit the necessary sampling. Even though several approaches have shown success, they often suffer from convergence issues, poor reproducibility, or are not well transferable to other systems.

In this work, we try to improve the convergence of the geometric route by scaling down the inter-molecular non-bonded interactions between the binding partners during the separation step. The unbinding free energy is calculated using an umbrella sampling replica exchange scheme, and contributions of the scaling are accounted for in separate, relatively inexpensive, free energy simulations. We could show that this method increased mixing in bottleneck regions by up to 15-fold in some test systems, considerably improving convergence. While the prediction of accurate binding free energies is still challenging, we suspect that this method can help to increase the feasibility of the geometric route for protein-protein systems by reducing the computational demand, thereby accelerating the development of novel protein-drugs.

P27 - Generative Active Learning for Molecular Design: Integrating Compound Synthesizability Prediction with Binding Affinity Optimization

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AstraZeneca

Generative AI combined with physics-based simulation offers a promising path to accelerated molecular design, yet practical drug discovery requires optimization of binding affinity, among other properties, while ensuring the compounds can be synthesized. Previously, we introduced a generative active learning (GAL) framework coupling the reinforcement learning platform Reinvent (a) with the ensemble-based binding free energy protocol ESMACS (b) as a binding affinity oracle. This approach demonstrated efficient iterative enrichment of strong binders across chemically diverse regions of space.(c)

Here, we extend this framework to a true multi-objective optimization problem by explicitly incorporating synthesizability constraints into the GAL cycle. We evaluate several different synthesizability scoring strategies. These scores are integrated directly into the reinforcement learning objective or are applied in a final post processing step. Two structurally and mechanistically distinct targets are investigated: TNKS2, featuring a confined binding pocket, and SARS-CoV-2 3CLpro, characterized by greater binding mode variability.

For TNKS2, we demonstrate that strong binders with high synthetic feasibility can be identified without substantial loss of binding affinity, depending on the chosen scoring scheme. Certain synthesizability models promote exploration of distinct chemical regions while maintaining high diversity among top candidates. In contrast, for 3CLpro, synthesizability constraints introduce a measurable trade-off, reducing achievable binding affinity but preserving structural diversity. Notably, applying synthesizability filtering only at the final iteration retains high affinity while maintaining synthetic tractability in case of TNKS2 but not necessarily for 3CLpro.

All large-scale simulations were performed on the Frontier exascale system. Our results highlight how the choice and timing of synthesizability constraints critically shape the Pareto balance between affinity, feasibility, and diversity, providing practical guidance for integrating generative AI with physics-based methods in real-world drug discovery workflows.

(a) H H Loeffler, J He, A Tibo, J P Janet, A Voronov, L H Mervin, O Engkvist, J. Cheminform., 2024, 16:20.

(b) S Wan, A P Bhati, S J Zasada, P V Coveney, Interface Focus, 2020, 10, 20200007.

(c) H H Loeffler, S Wan, M Klähn, A P Bhati, P V Coveney, J. Chem. Theory and Comput., 2024, 20, 8308.

P28 - Relative Binding Free Energy pipeline for G Protein Coupled Receptors

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IMEC

G protein-coupled receptors (GPCRs) are proteins located in the cell membrane, and they are common targets in drug development. The setup of simulations for them is more complex than that for soluble globular proteins due to the incorporation of the lipid membrane, larger solvent box and deep pockets where water molecules get trapped.

We have developed an open source relative binding free energy (RBFE) pipeline based on the paper by Moore et al. (Moore, 2023) and extended it to simulate GPCRs. The input files needed to run our pipeline are ligands' SMILES, coordinates of the docked reference ligand and a PDB file of the equilibrated system of the protein - reference ligand surrounded by the membrane.

The pipeline uses the python packages RdKit and pmx for generating 3D molecules and creating hybrid topologies, respectively. AutoDock Vina is used to dock the ligands and LOMAP to create free energy perturbation (FEP) graphs.

We use the non-equilibrium approach for the simulations and $\Delta\Delta G$ values for each edge are obtained using the Crooks fluctuation theorem (with pmx analyze). MD simulations are run using GROMACS. Before calculating absolute binding free energies, cycle closure corrections are applied to obtain the best free energy estimators in FEP paths.

We tested the pipeline on a set of 10 ligands that bind to the A2A GPCR and achieved an accuracy of ~ 1 kcal/mol relative to experimental binding free energies. The workflow is easy to use and customize for other protein-ligand systems. The simulation times and number of transitions desired per system can be easily modified by the user.

References

Moore, J. H. (2023). Automated relative binding free energy calculations from SMILES to $\Delta\Delta G$. *Communications Chemistry* 6, 82.

P29 - From Unbiased MD to Selectivity: Neural-Network Metastable State Discovery and MSM Kinetics for Bioorthogonal Analogs

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Metabolic glycoengineering uses “clickable” nucleotide-sugar donor analogs, but their efficacy and selectivity often depend on the incorporated bioorthogonal tags, which can shift binding-mode populations by altering bound-state dynamics, effects that conventional structural or binding analyses may fail to quantify. Here we present a workflow that links conventional, unbiased molecular dynamics (MD) to kinetic modeling to make quantitative predictions of donor selectivity for wild-type (WT) versus engineered “bump-and-hole” (BH) N-acetylglucosaminyltransferase I (MGAT1) variant.

We run a series of MD simulations for each enzyme–donor complex and analyze the trajectories with a neural-network dimensionality-reduction approach (VAMPnets) trained on time-lagged features extracted from the trajectories to capture the slow binding dynamics. This yields a low-dimensional description of binding modes and separates the simulations into a small number of metastable states. We then build Markov state models (MSMs) to compute equilibrium populations and transition kinetics between these states. In practice, this lets us quantify the balance between two pose ensembles interpreted as “productive” and “nonproductive” Michaelis complexes, and how quickly the system moves between them. Overall, combining VAMPnets with MSM kinetics makes MD more predictive: we estimate how likely catalytically ready poses are populated and how stable they are over time, enabling quantitative selectivity predictions. More broadly, this approach links metastable state populations and interconversion kinetics to two practical design paths: guiding BH enzyme engineering by identifying mutations that favors productive populations for preselected donor candidates while depleting nonproductive states (optimizing enzymes for preselected donors), and guiding clickable-donor design by prioritizing analogs that preferentially stabilize productive binding ensembles in the target enzyme (optimizing donors for preselected enzymes). In general, this approach can help predict engineering strategies to enhance catalytic efficiency by identifying mutations that favors productive populations, guide specificity tuning by revealing modifications that favor certain substrates and enable both the design and prediction of molecules that are selective for a target protein or variant.

P30 - Comparing co-alchemical ion and analytic correction charge-changing perturbation strategies for modeling protein mutations in λ -dynamics

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Charge-changing perturbations are notoriously difficult to investigate with alchemical free energy (AFE) calculations. Periodic boundary conditions and electrostatic approximations, which are routinely employed in AFE calculations, introduce finite-size effect errors that become non-negligible when the simulation cell's net charge is perturbed away from zero. Two prevalent strategies exist to correct for these errors: the analytic correction (AC) and co-alchemical ion (CI) methods. Although both correction schemes have been shown to produce comparable relative free energy results for small molecule perturbations, a side-by-side comparison of these methods has not been performed for protein side chain mutations or for use with λ -dynamics (λ D) free energy calculations. The use of the CI approach with λ D has also been limited by a lack of infrastructure to assist users with its more involved setup procedure. This work introduces an automated implementation workflow for the CI approach with λ D and then compares the performance of these two charge-change correction schemes. Specifically, the relative folding and binding free energies of a series of charge-changing perturbations of EXOSC3 (D132A, G135R, and G191D) were performed with λ D using both the CI and a simplified AC correction schemes. Excellent agreement between simplified AC- and CI-corrected free energy results was observed, with a mean unsigned error of 0.4 kcal/mol. However, reduced sampling proficiency and increased convergence difficulties for multisite perturbations were observed for the CI method, suggesting that a simplified AC approach may be more generalizable for future λ D calculations.

P31 - Site-directed mutagenesis applying A-EDS

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Amino acid mutations can modulate protein stability and ligand binding affinity by reshaping local interactions and conformational ensembles. Standard alchemical free energy methods (e.g., FEP/TI) typically require pairwise simulations per mutation, which scales poorly for a large set of mutations. An attractive alternative to classical alchemical perturbation methods is accelerated enveloping distribution sampling (A-EDS) [1,2]. This is achieved by using a reference Hamiltonian, which combines the Hamiltonians of the end-states, weighted according to their respective Boltzmann factors. A-EDS is already successfully applied for the calculation of free-energy differences between multiple ligands from a single simulation [3].

Here, we assess the applicability of A-EDS to site-directed mutagenesis, targeting both protein stability and protein–protein binding in the trypsin–BPTI complex. In a first step different combinations of amino acids were tested to ensure balanced sampling across amino acid variants in the solvent. We then apply two modes within the protein context: a screening mode that classifies mutations as ‘beneficial’ or ‘deleterious’, and a quantitative mode where the parameters are adapted to the local environment to obtain free-energy differences.

REFERENCES

- (1) Christ, C. D.; van Gunsteren, W. F. Enveloping distribution sampling: A method to calculate free energy differences from a single simulation. *The Journal of Chemical Physics* 2007, 126, 184110.
- (2) Perthold, J. W.; Petrov, D.; Oostenbrink, C. Toward Automated Free Energy Calculation with Accelerated Enveloping Distribution Sampling (A-EDS). *Journal of Chemical Information and Modeling* 2020, 60, 5395–5406, PMID: 32492343.
- (3) O. Gracia Carmona, M. Gillhofer, L. Tomasiak, A. de Ruiter, C. Oostenbrink Accelerated enveloping distribution sampling to probe the presence of water molecules. *J. Chem. Theory Comput.* 19 (2023) 3379 – 3390.

P32 -

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In small-molecule drug discovery, accurate and efficient prediction of protein–ligand interactions is essential for identifying leads for new targets, yet current AI models are limited by the lack of large, high-quality structural and binding data, which restricts their generalization in real-world settings. Physics-based high-performance computing (HPC) helps overcome this limitation by generating large, high-fidelity datasets for training and validating advanced AI models, enabling exploration of molecular interaction space at unprecedented scale.

Leveraging this paradigm, BindingNet v2 was constructed as an extensively expanded dataset of 689,796 modeled protein–ligand complexes across 1,794 protein targets, built via an enhanced template-based workflow from BindingNet v1 that integrates advanced pharmacophore and molecular shape similarity methods to improve binding mode prediction. When used to augment the PDBbind dataset, BindingNet v2 markedly improved the Uni-Mol model’s pose prediction performance on the challenging PoseBusters benchmark, increasing the success rate from 38.55% to 64.25%, underscoring the importance of large, diverse datasets for robust AI in drug discovery.

Additionally, machine learning interatomic potentials (MLIPs) enable quantum-level molecular modeling but require robust training data. PANIP, an ensemble model built on NequIP, uses a multi-fidelity active learning workflow to extract diverse, information-rich subsets, focusing expensive QM calculations on key structures. Trained on non-covalent protein interactions, PANIP compresses data to 8.7% while maintaining accuracy (MAE < 0.2 kcal/mol). It outperforms ANI 2x, generalizes well to biomolecular systems, and achieves near-force-field speed with QM precision.

More broadly, the integration of HPC and AI is enabling more accurate and efficient calculations of molecular interactions, offering a path to systematically address data scarcity and enhance predictive power. As AI architectures are increasingly co-designed for HPC environments, molecular interaction predictions can be performed more rapidly and at lower cost, promising to accelerate drug discovery timelines, reduce development expenses, and deliver new therapies to patients more quickly and effectively.

P33 - Rational design of novel pyrimidinone derivatives as dual A2A/A2B adenosine receptor antagonists

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Over the last decade, immune-oncology therapies have achieved breakthrough records of survival rates for several types of cancer. However, they face a challenge in the presence of multiple metabolic dysregulations in the tumor microenvironment that promote immunosuppression. One of such is the high concentration of adenosine maintained due to the hydrolysis of ATP released by cells in conditions of hypoxia, which then binds to adenosine receptors expressed on effector T and NK cells to exert an immunosuppressive response. As such, adenosine receptors (in particular A2A and A2B) have emerged as a promising drug target for cancer immunotherapy, with several molecules on clinical trials. In this work we characterize a novel series of compounds derived from a family of selective A2B adenosine receptor antagonists found through our drug discovery program. These derivatives are instead dual A2A/A2B antagonists developed as potential lead candidates to improve lymphocyte activity. We conducted a docking exploration of these compounds on adenosine A2A and A2B receptors and found a potential dual binding mode, which was then interrogated by systematic free energy perturbation (FEP) calculations with QligFEP. While FEP finds better agreement with the conserved binding mode proposed for the parent A2B selective pyrimidinone compounds, it opens the door to an alternative binding orientation that might be relevant to explain the dual A2A/A2B antagonism. These results are guiding the design of new compounds that are undergoing experimental validation.

P34 - Multisite λ Dynamics in Academic Drug Design Projects

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Over the last four years we have used multisite λ dynamics (MS λ D) for calculating protein-ligand relative binding free energies in our academic projects. The projects involved three different targets of pharmaceutical relevance with varying but interconnected goals.

First, we explored the use of MS λ D in retrospective and prospective affinity predictions for Human Neutrophil Elastase (HNE) and Proteinase 3 (PR3) respectively, both drug targets for chronic lung inflammation. Four different force field combinations were evaluated for retrospective predictions from which the hybrid CHARMM36m-OPLS-AA force field (FF) predicted highest correlation. The prospective predictions for PR3 with this hybrid FF revealed the polypharmacological potential of the existing HNE inhibitors.

Next, we proposed a λ -dynamics based methodology for accurately ranking alternate ligand binding poses. We tested the validity and predictive power of our approach using two pharmaceutically relevant targets (HNE and Leishmania major N-myristoyltransferase) and eight compounds from experimentally characterized congeneric series. For each target, our approach correctly ranked the known X-ray poses to be more favorable than alternative flipped poses.

Currently, we are applying MS λ D and relative binding free energy calculations in a de-novo drug design setting for targeting the FabF enzyme, a novel target for new antibiotics.

P35 - Scalable, accurate, and adaptive free energy calculations for molecular design

Lindsey Whitmore, Satya Pratik Srivastava, Hattie Barber, Antonia Mey, Vincent Voelz, Michael Shirts

University of Colorado Boulder

Relative binding free energy (RBFE) calculations offer high predictive accuracy for small molecule potency ranking, but scaling these methods across the large ligand networks encountered in lead optimization remains a significant practical challenge. We present a scalable RBFE platform integrating the OpenFreeEnergy (OpenFE) and Alchemiscale frameworks with Folding@Home distributed computing infrastructure, enabling simultaneous execution of free energy calculations across hundreds of candidate ligands. Using the TYK2 kinase system as an initial benchmark, we validate our pipeline and establish the convergence and uncertainty diagnostics needed to make reliable, network-level binding affinity predictions.

Building on this foundation, we are integrating active learning to close the loop between simulation results and experimental prioritization. Rather than exhaustively sampling a full perturbation network, our framework uses current model uncertainty and predicted information gain to dynamically select which ligand transformations to run next, focusing compute resources where they will most improve decision-making. By coupling scalable free energy execution with intelligent sampling, we aim to significantly reduce the number of simulations required to confidently rank a lead series.

P36 - Non-equilibrium alchemical thermodynamic integration to calculate accurate free energy differences in biomolecular systems

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Computational protein and drug design benefit greatly from accurate free energy difference estimations upon amino acid mutation or ligand perturbation. Empirically derived score functions provide fast free energy difference estimates, but often lack in precision and accuracy, especially at early stages of drug development, where the chemical space is vast and data are scarce. Alchemical methods based on molecular dynamics (MD), such as free energy perturbation (FEP)—the most common implementation of alchemical calculations—are more computationally expensive, but have shown the required level of performance. Here, we take a comprehensive look at non-equilibrium alchemical thermodynamic integration (TI), which calculates the free energy difference by integrating the derivative of the Hamiltonian with respect to a λ coupling parameter between end states along a non-equilibrium, non-physical transformation. We show how non-equilibrium TI can be incorporated into ligand design and mutant screening using open-source software, achieving accuracies comparable to more popular methods at a lower computational cost.

Poster Session 2

P37 - Stabilizing Large Alchemical Perturbations in Relative Binding Free Energy Calculations

Donald van Pinxteren and Willem Jespers

University of Groningen

Relative binding free energy (RBFE) calculations are now routinely applied in lead optimization campaigns, where they can provide actionable guidance at chemical accuracy. Despite these successes, the broader applicability of RBFE remains constrained by well-known methodological limitations related to sampling efficiency, numerical stability, and uncertainty quantification. QligFEP v2.0, an automated RBFE workflow developed by Araripe et al., significantly lowered the barrier to applying alchemical free energy methods by standardizing setup procedures (Araripe et al., 2025). However, these workflows rely on transformation schemes, where perturbation sizes between ligands must remain small to ensure sufficient phase-space overlap. In practical drug discovery settings, phase-space overlap requirements limit chemical exploration, as realistic optimization campaigns frequently involve modifications that exceed the comfortable range of traditional RBFE protocols. Increasing perturbation sizes is therefore essential to improve throughput, and allow RBFE to operate on chemically more diverse compound series. Poorly behaved alchemical intermediates or insufficient phase-space overlap can destabilize simulations, resulting in inflated statistical uncertainties that compromise the reliability of free energy estimates (van Pinxteren & Jespers, 2025). In current RBFE practice, maintaining stability under increasingly challenging perturbations often requires dense λ -schemes and extensive sampling, substantially increasing computational cost (Baumann et al., 2023). For this reason, a virtual particle restraint method is explored within QligFEP v2.0 to stabilize MD simulations for larger alchemical changes, initially focusing on ring transformations. In this approach, a virtual particle is placed at the center of corresponding ring systems in both ligands and restrained relative to each other without directly influencing atomic behavior. Preliminary results indicate a substantial reduction of the standard error of the mean over 10 replicate simulations for solvation free energy calculations.

Shen Guo, Chiel Jespers, David Alencar Araripe, Willem Jespers

Groningen University

GPU acceleration has become a key strategy for improving the performance of molecular dynamics (MD) simulations, enabling efficient exploitation of parallelism in particle-based computation[2]. QGPU is a GPU-accelerated adaptation of Q version 5.06, designed as a lightweight molecular dynamics (MD) engine with a focus on high computational efficiency. The original Q engine employs spherical boundary conditions to restrict simulations to a localized region, enabling efficient sampling while maintaining accuracy[1]. Building on this design, QGPU offloads the majority of computational workloads to the GPU and introduces optimized kernels for non-bonded interactions, significantly improving performance.

Due to the relatively small system sizes enabled by spherical boundary conditions, QGPU is able to efficiently execute multiple simulation replicates concurrently on a single GPU. Benchmark results show that, when running 10 concurrent replicates, QGPU achieves up to 200× speedup compared to the CPU-based implementation on A100. This demonstrates the advantage of GPU-based parallelism for high-throughput MD simulations in confined systems.

QGPU thus provides an efficient platform for large-scale sampling through concurrent simulations. Future work will focus on further performance optimization, integration with qligfep[3], and support for a broader range of force fields.

[1] Bauer P., Barrozo A., Purg M., Amrein B. A., Esguerra M., Wilson P., Major D. T., Åqvist J., Kamerlin S. C. L. (2018). Q6: A comprehensive toolkit for empirical valence bond and related free energy calculations. *SoftwareX* 7, 388–395. URL: <https://www.sciencedirect.com/science/article/pii/S2352711017300675>, doi:<https://doi.org/10.1016/j.softx.2017.12.001>.

[2] Stone J. E., Hardy D. J., Ufimtsev I. S., Schulten K. (2010). Gpu-accelerated molecular modeling coming of age. *Journal of Molecular Graphics and Modelling* 29(2), 116–125. URL: <https://www.sciencedirect.com/science/article/pii/S1093326310000914>, doi:<https://doi.org/10.1016/j.jmgm.2010.06.010>.

[3] Jespers, W., Esguerra, M., Åqvist, J., & Gutiérrez-de-Terán, H. (2019). QligFEP: an automated workflow for small molecule free energy calculations in Q. *Journal of cheminformatics*, 11(1), 26. <https://doi.org/10.1186/s13321-019-0348-5>

P39 - Fast GPCR Ligand RBFE in All-Atom Lipid Bilayers

David Alencar Araripe, Wessel Porschen, Willem Jespers

University of Groningen

G protein-coupled receptors (GPCRs) are a major drug target family. Free Energy Perturbation (FEP) provides rigorous estimates of relative binding affinities for lead optimization, yet applying FEP to GPCRs presents a unique challenge: their transmembrane nature demands modeling the lipid bilayer, which increases system complexity and computational cost. Previous FEP studies on these GPCR targets used either Q with spherical boundary conditions (SBC) and united-atom (UA) Berger lipid parameters, or periodic boundary conditions (PBC) paired with all-atom force fields. While the SBC approach with UA lipids yields stable simulations, it may lack the atomic detail to fully capture lipid-protein and lipid-ligand interactions.

QligFEPv2 is an open-source workflow for relative binding free energy calculations using the Q molecular dynamics engine. Recently benchmarked against commercial and open-source methods, QligFEP achieved competitive accuracy at an accessible computational cost. While the combination of Amber14sb and Lipid21 was used in PBC simulations, to our knowledge, our work represents the first application of SBC FEP on GPCRs using an AA lipid force field. This is achieved through an OpenMM-based protocol that places Lipid21 POPC bilayers around the protein structure, replacing the previous GROMACS-based UA approach. GPU-accelerated relaxation optimizes lipid packing in ~9 min per system (V100). Ligand poses are generated via maximum common substructure (MCS)-constrained conformer generation, followed by protein-aware flexible alignment to orient substituents within the binding pocket.

We apply this approach to multiple GPCR targets, encompassing over 120 ligands parametrized with OpenFF Sage 2.3. Results across the benchmark set show statistically significant rank-order correlation with experimental binding affinities for the majority of targets (7/12; Benjamini-Hochberg corrected), with per-edge replicate runtimes of approximately 3 h on a 16 CPU cores.

P40 - TACTICS: Bayesian Active Learning for Compound Prioritization in Ultra-Large Combinatorial Libraries Toward Free Energy Calculations

Remco L. van den Broek; A. Nandkeolyar; Donald J. M. van Pinxteren; G. J. P. van Westen; D. L. Mobley; W. Jespers

LACDR | Leiden University

De novo molecular design increasingly faces a translation bottleneck: while generative models can propose vast numbers of candidates, scalable synthesis remains constrained. Combinatorial synthesis offers a practical alternative by enabling rapid construction and screening of millions of compounds from a limited set of building blocks¹. However, this paradigm introduces a new challenge: efficiently sampling an exponentially large combinatorial space to identify high-value compounds.

Here, we present TACTICS, a flexible Bayesian framework for efficient sampling of ultra-large combinatorial libraries. Rather than relying on random selection or exhaustive enumeration, TACTICS prioritizes combinations that maximize expected information gain. The framework supports previously published Thompson Sampling workflows^{2,3}, and extends them with a higher-efficiency strategy tailored to diverse library compositions and scoring functions. This improved strategy employs Top-Two Thompson Sampling (TT-TS) for enhanced best-arm identification, combined with adaptive per-component thermal cycling and GMIC-weighted component rotation to balance exploration and prioritize critical candidates. By iteratively updating its estimates, TACTICS focuses sampling on components most likely to contribute to binding. These enhancements improve top-100 recovery (76% → 86.1%), defined as the fraction of ground-truth top-100 compounds identified during sampling, compared to the current-best published Thompson Sampling workflow³. This result is averaged across 21 benchmark libraries constructed using ROCS similarity or docking scores. Performance gains are achieved without increasing the call budget, enabling more effective identification of enriched building blocks than existing approaches.

Importantly, TACTICS is designed to operate upstream of high-accuracy, computationally intensive methods such as alchemical free energy calculations, directly addressing the translation bottleneck between large-scale molecular generation and rigorous evaluation. By applying Thompson Sampling to ultra-large combinatorial libraries, TACTICS identifies a focused subset of candidates that are both enriched for high-affinity binders and structurally coherent. Restricting free energy perturbation (FEP) calculations to this subset substantially reduces redundancy while enabling the construction of well-connected perturbation networks through shared building blocks. This connectivity promotes chemically consistent transformations, improving the stability and interpretability of FEP results. Overall, TACTICS enables the systematic reduction of vast chemical spaces into tractable, high-quality candidate sets, facilitating efficient and reliable downstream free energy workflows.

(1) van der Nol, E.; Haupt, N. A.; Gao, Q. Q.; et al. Barcode-free Hit Discovery from Massive Libraries Enabled by Automated Small Molecule Structure Annotation. *Nat. Commun.* 2025, 16, 9479. <https://doi.org/10.1038/s41467-025-65282-1>.

(2) Klarich, K.; Goldman, B.; Kramer, T.; et al. Thompson Sampling—An Efficient Method for Searching Ultralarge Synthesis on Demand Databases. *J. Chem. Inf. Model.* 2024, 64 (4), 1158–1171. <https://doi.org/10.1021/acs.jcim.3c01790>.

(3) Zhao, H.; Nittinger, E.; Yu, M. A.; et al. Enhanced Thompson Sampling by Roulette Wheel Selection for Screening Ultralarge Combinatorial Libraries. *J. Cheminform.* 2025, 17, 154. <https://doi.org/10.1186/s13321-025-01105-1>

P41 - Validating peptide-probe unbinding with contact parallel cascade selection molecular dynamics (cPaCS-MD) for fluorescent-guided surgery

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Leiden University Medical Center

Fluorescence-guided surgery (FGS) is a promising and rapidly growing field, where fluorescence imaging offers real-time intraoperative guidance, promoting favorable treatment prognosis.(1) Peptide-based probes are of great interest, providing high affinity and good specificity to targets, while maintaining relatively cheap production costs.(2, 3) Recently, several deep learning-based peptide binder generators, such as RFpeptides and BindCraft, have been developed to rapidly generate large libraries of candidate binders, which are subsequently filtered using confidence scores and/or low-cost physics-based metrics.(4, 5) However, fluorescence-guided screening (FGS) typically requires the conjugation of a bulky and potentially charged dye to the peptide, which can significantly influence binding interactions. Additionally, the costs of dye labeling limits the number of peptide-probe candidates that can be feasibly synthesized and tested. This study will evaluate the application of contact parallel cascade selection molecular dynamics (cPaCS-MD) for downstream peptide-probe unbinding path and energy prediction in peptide-probe design workflows.(6) Folate receptor alpha (FR α), a promising protein biomarker for FGS, will be targeted with RFpeptides to design novel cyclic peptide binders. The top macrocycles will be modified with Indocyanine green (ICG), a cyanine dye commonly used in FGS. cPaCS-MD will simulate the constructed macrocycle-probes unbinding path and (un)binding energy within the appropriate tumor environment to select the most promising macrocycle-probes for FGS. RFpeptides selected 20 peptides based on iPAE, rosetta affinity prediction, spatial aggregation propensity scores, and contact molecular surface, which were consequently analyzed for predicted binding affinity and rescored with cPaCS-MD. The macrocycle-probes will be synthesized and evaluated in vitro for binding affinity using SPR and for FGS applicability with Life Cell Imaging. This study aims to establish the utility of cPaCS-MD as a predictive tool in peptide binder design and to evaluate its capacity to predict binding affinity changes resulting from probe conjugation in the context of FGS.

P42 - An integrated residue and ligand free energy perturbation protocol for membrane proteins

Wessel Porschen, Willem Jespers, Hugo Gutiérrez-de-Terán

MODSIM Pharma AB / Research Center of Nanotechnology and Nanomaterials (CINN/CSIC) / Leiden University

At MODSIM Pharma, we specialize in Structure-Based Drug Design with our physics-based methods and molecular simulations to optimize Drug Discovery. One of our main areas is the development of modulators of G-protein Coupled Receptors (GPCRs) and other membrane receptors, taking into consideration that these families of proteins cluster more than 70% of all therapeutic targets for current drugs in the market. We present here our unified pipeline for membrane-receptor lead optimization, based on a combination of modeling and simulation of receptors in an atomistic representation of the membrane environment, with our efficient spherical-boundary representation of the binding site for free energy perturbation simulations. The first stage combines PyModSym and PyMemdyn, for effective modeling and MD simulations of any membrane receptor including quaternary complexes with associated proteins and/or ligands of any kind, to produce a membrane-embedded, solvated, and equilibrated system, ready to be used as a starting point for further MD simulations, including ligand-binding free energy calculations. The systems are then transferred to the spherical boundary conditions (SBC) of our Q-FEP protocols, ready to perform accurate and scalable ligand free energy calculations focusing on the binding site. On the one hand, QligFEP v2.1.0 is a flexible open-source workflow based on a graphical and command-line interface for calculating relative binding free energies, and automatically handles diverse chemical transformations, streamlined setup procedures, and enhanced analysis tools. On the other hand, QresFEP-2 is a novel hybrid-topology FEP protocol benchmarked on a comprehensive protein stability dataset, as well as protein-protein interactions and protein-ligand binding datasets. Recent illustrative examples of our toolbox include the elucidation of the correct experimental binding orientation of an A2BAR Partial Agonist or the identification of CCR2 allosteric ligands.

P43 - QmapFEP - A flexible infrastructure for high-throughput FEP calculations with spheric boundary conditions

Chiel Jespers, Joshua Tromp, David Araripe, Willem Jespers

Leiden University

Spherical boundary conditions (SBC) represent an appealing way of reducing computational costs of FEP calculations by only requiring simulations for a small region around the binding site. In previous work the automated platform QligFEP for the Q molecular dynamics package[1] was introduced that made such calculations available for high-throughput applications[2], but adapting resource management to a wide variety of available server architectures remains a challenge.

The present work delivers a flexible SaaS solution called QmapFEP that can directly operate on slurm clusters, requiring only ssh access to set up a connection. Another architecture that can be targeted is the Microsoft Azure cloud, allowing users to scale up more computational resources on the spot. The automated workflow of QligFEP is exposed as a unified portal through which the progress of the simulation can be monitored and results can be viewed and analyzed.

As a case study the architecture is applied to a cluster of Raspberry Pi computers on which a distributed slurm cluster was installed automatically, highlighting the efficiency of the solution in terms of computational costs as well as the ease of use.

[1] Q: A Molecular Dynamics Program for Free Energy Calculations and Empirical Valence Bond Simulations in Biomolecular Systems. Marelus, J. et al. J. Mol. 422 Graph. Model. 1998, 16 (4-6), 213–225

[2] Doing More with Less: Accurate and Scalable Ligand Free Energy Calculations by Focusing on the Binding Site. D. Araripe et al. Journal of Chemical Information and Modeling 2026 66 (6), 3164-3172

P44 - Applications for Alchemical Free Energy Calculations of Solvation, Binding and pKa in Complex Biomolecular Systems

Qinghua Liao, Birgit Strodel, Lynn Kamerlin, Carme Rovira

University of Barcelona

Alchemical free energy calculations have been widely used to tackle different biological challenges. In this presentation, we illustrate a broad range of applications from dummy model development of metal ions to substrate recognition and protonate state of catalytic residues of enzymes.

We developed a nonbonded Cu^{2+} model that includes the Jahn–Teller effect (Liao et al. JPCL, 2015), as well as an extension of the nonbonded cationic dummy model to capture ion-induced dipole interactions (Liao et al. JPCL, 2017). These models were parameterized by reproducing experimental hydration free energies and radius distribution function, and were further validated in different complex biological systems.

Moving from solvation to molecular recognition, we investigate a unique family of di-glutamate mucinases that evolutionarily bridges glycosyl hydrolases and peptidases. We utilized alchemical methods to compute the absolute binding free energy of the N-acetylgalactosamine (GalNAc) moiety to the enzyme. These calculations reveal the atomistic insights of substrate specificity (Narimatsu et al. Nat. Catal. 2024)

Finally, we apply these methods to probe the catalytic machinery of a T6SS toxin. By calculating the pKa shift of a critical lysine in the active site, we demonstrate that this residue maintains neutrally charged under physiological conditions. This evidence confirms the lysine's role as a general acid/base, effectively resolving a key step in the toxin's mechanism of action (in preparation 2026). Together, these studies highlight the versatility of alchemical methods in solving different challenges across computational biochemistry.

P45 - Alchemical free energy calculations with neural network potentials

Sara Tkaczyk, Thierry Langer, Marcus Wieder, Andrea Rizzi, Stefan Boresch

University of Vienna

Neural network potentials (NNPs) have emerged as a promising alternative to conventional force fields for molecular simulations, offering near-quantum chemical accuracy at significantly reduced computational cost. Recently, we demonstrated the feasibility of performing alchemical free energy calculations using pretrained, transferable NNPs within a single-coordinate dual-topology framework in explicit water. In this approach, alchemical transformations are realized without altering the NNP architecture, using energy mixing and neighborlist modifications to mask dummy atom interactions. Building on this foundation, we now explore methodological refinements and extensions of this framework with the aim of improving accuracy and/or efficiency in condensed phase free energy simulations. In particular, we focus on tautomer ratios in aqueous solution as a model system. Using the pretrained MACE-OFF potentials, we investigate fully NNP-based alchemical simulations as well as hybrid schemes that combine NNP simulations with MM descriptions, and we explore combining QM calculations with MD-based free energy simulations to determine free energy differences. Furthermore, we assess how this framework could be extended to larger and more complex alchemical transformations.

P46 - Optimizing and Applying LaDyBUGS for Accurate Modeling of Protein-Peptide Binding Free Energies

Sarah Stieglitz, Jonah Vilseck

Indiana University School of Medicine

Peptide-based therapeutics are an emerging area of drug development, capable of precisely modulating protein-peptide interactions. Single amino acid substitutions within a peptide can dramatically alter binding affinity, influencing disease progression or enabling therapeutic design. While alchemical free energy (AFE) methods are powerful tools for computing free energy changes associated with chemical mutations, their application has largely centered on small molecule design, with limited emphasis on peptides. Thus, this research aims to optimize an AFE framework to efficiently model amino acid mutations in peptide-protein complexes and advance peptide-based therapeutic development. Specifically, the λ -Dynamics with Bias-Updated Gibbs Sampling (LaDyBUGS) AFE method is evaluated for its ability to accurately reproduce experimental binding data accompanying single amino acid mutations in six protein-peptide complexes selected from the SKEMPI 2.0 database.

In one representative system examined to date, LaDyBUGS has demonstrated encouraging agreement with experimental data, particularly for mutations to uncharged residues. Across 36 mutations, a root mean square error (RMSE) of 2.24 kcal/mol and Kendall τ rank correlation of 0.78 were obtained. However, mutations to charged residues showed larger $\Delta\Delta G_{\text{bind}}$ discrepancies from experiment. Structural analysis revealed that the mutated positions on the peptide lie within a hydrophobic pocket, where charged substitutions would be energetically unfavorable and titratable side chain may feature shifts in their pKa values. Incorporating a pKa correction scheme substantially improved LaDyBUGS $\Delta\Delta G_{\text{bind}}$ agreement with experiment, decreasing the overall RMSE to 1.23 kcal/mol while maintaining a strong τ correlation of 0.77. Overall, these results demonstrate that LaDyBUGS, when paired with pKa and protonation state correction factors, is a promising AFE approach for predicting mutation-based binding affinity changes in protein-peptide systems.

P47 - Smoother Alchemical Transformation via Enveloping Distribution Sampling for Free-Energy Estimation

Shu-Yu Chen, Enrico Ruijsenaars, Philippe H. Hünenberger, and Sereina Riniker

University of Basel

The accuracy of the computational estimation of relative free energies (e.g., for solvation or protein-ligand binding) depends on the smoothness of the phase-space transformation between the two alchemical end states. A smooth transformation ensures sufficient phase-space overlap between the neighboring intermediate states connecting the two end states in equilibrium (EQ) simulations and generates less dissipative work in non-equilibrium (NEQ) simulations. The conventional energy interpolation (EI) coupling scheme constructs the intermediate states by linearly combining the end-state potentials. We show that the enveloping distribution sampling (EDS) coupling scheme, a generalization of EI where the corresponding Boltzmann factors are linearly combined, represents a much more flexible alternative. Across a range of systems, from 1D harmonic oscillators and 2D Ising models to the absolute hydration free energies of small molecules, the EDS coupling scheme consistently yields more accurate and statistically robust free-energy estimates compared to the conventional EI scheme in both EQ and NEQ simulations.

P48 - Fast, accurate prediction of protein-ligand binding free energies by mining minima: the VM2 software package

Simon P. Webb, Michael J. Potter, Michael K. Gilson, Lawrence E. Stewart

VeraChem LLC

VeraChem's software package VM2, uses the second-generation mining minima statistical mechanics method to calculate protein-ligand binding free energies. Benchmarking studies have shown that VM2 can provide similar ligand series rank ordering accuracy to free energy perturbation methods, but it takes only 10-20 minutes per protein-ligand calculation on CPUs, with further speed up on GPUs. Binding free energy calculations with VM2 include controllable levels of conformational searching, entropic effects, bulk solvation effects through various continuum models, and explicit water molecules in the binding site when needed. VM2 is an end-point method, independently calculating the chemical potentials of the ligand-protein complexes and the unbound ligands and protein to provide the binding free energy of each ligand in a series. This allows for very high scaling of ligand series throughput across compute clusters, and it also makes the method amenable to corrections with quantum mechanical (QM) potentials for more reliable accuracy as well allowing application to covalent protein-ligand binding. This talk will describe the VM2 methodology and the end-to-end workflow that provides automated protein-ligand series system preparation, initial posing of ligands, calculation submission, and results collection. Results of VM2 benchmark studies against experiment will be presented, including with QM corrected binding free energies. Furthermore, ongoing work to extend the applicability of VM2 from lead-optimization to earlier stages of preclinical drug discovery will be included, for example, VM2-based high-throughput screening to rank hits provided by conventional docking or deep learning methods. VM2's interface to Chemical Computing Group's integrated molecular modeling platform MOE will also be described.

P49 - Accelerating Molecular Dynamics

Yutong Zhao, Mario Geiger, Yuxing Peng, Emine Kucukbenli

NVIDIA

Making molecular dynamics fast and flexible on commodity hardware (eg. GPUs) has been a long standing challenge for the development of free energy methods. In the age of AI-tuned hardware and ML frameworks, we'd like to present our general approach and thinking towards accelerating primitives in molecular dynamics without sacrificing ease of use.

P50 - Exploring the Energetic Contributions of Halogen Atoms to Protein-Ligand Interactions

Zhao Chen, Yuanxun Wang, Niu Huang

National Institute of Biological Sciences, Beijing

Halogen atoms are highly valued in drug discovery due to their lipophilicity and their potential to form halogen bond (XB) interactions. However, the limited understanding of the factors that govern the formation of favorable XBs in protein-ligand interactions highlights the need for a thorough investigation of their structural and energetic properties. In addition, with the development of various computational tools for designing favorable XBs during lead optimization, a systematic benchmarking of these methods for calculating halogen interaction energies in protein-ligand complexes is essential. In this study, we constructed a data set of protein-ligand complex structures featuring halogen interactions and evaluated the performance of molecular mechanics (MM) and quantum mechanics (QM) methods in predicting the relative binding affinity. QM achieved the best performance, with Pearson correlation coefficient of 0.921 for XB systems and 0.667 for hydrophobic interaction systems. In contrast, MM methods were less effective because of inaccuracies in XB energy terms and their failure to account for polarization and induction effects. We also performed a detailed analysis of factors affecting performance, including ligand binding poses, desolvation effects, and the complex energy components and electron distributions associated with halogen atoms. By integrating MM and QM methods, we developed an efficient structure-based XB 'design in' workflow for lead optimization. The practical value of this workflow was demonstrated by designing XB interactions in kinase targets.

P51 - Free Energy Calculations in Fragment-Based Drug Discovery: Learnings from Internal Validation Studies

Pavel Buslaev, Barbara Farkaš, Carmine M. Morgillo, Chris Hamlett, Davide Branduardi, Paul N. Mortenson

Astex Pharmaceuticals

Astex has pioneered fragment-based drug discovery (FBDD) over the last 25 years and has an unrivalled collection of over 19,000 protein-ligand complex structures accompanied by high-quality experimentally determined affinity data spanning diverse target families. This provides a unique opportunity for testing various computational binding affinity prediction methods.

Here we highlight several internal studies of binding free energy methodologies, including large-scale relative binding free energy (RBFEP) studies and application of absolute binding free energy (ABFEP) methodology to a rich internal dataset of protein-fragment structures.

To study whether ABFEP is applicable to fragments, we compiled a dataset of internal fragment screening campaigns featuring high-resolution structures of all encountered hit matter and employed automated ABFEP calculations to predict corresponding experimental binding affinity indicators. ABFEP has shown satisfactory correlation with a range of experimental techniques covering both biophysical and enzymatic assays, successfully ranking diverse scaffolds, and distinguishing the dominant binding pose when multiple ones were experimentally identified over all systems under investigation. In our hands, ABFEP demonstrated strong performance for fragments and can be a strong supporting technology to orthogonally validate fragment hits with the potential to guide virtual screens and fast-track early stage FBDD.

P52 - TOWARDS ACCURATE BINDING MODE PREDICTION FOR FREE-ENERGY CALCULATIONS: DEVELOPMENT OF THE MULTIPLE-COPIES ASSOCIATION STUDIES (MAS)

Varbina Ivanova, Patricia Blanco-Gabella, Álvaro Serrano, Xavier Barril, Jordi Juárez-Jiménez

University of Barcelona

Accurate prediction of protein–ligand binding free energies relies critically on the quality of the initial binding mode used for the calculations. However, when structural data is limited or chemical modifications introduce alternative ligand orientations, selecting a reliable binding pose becomes challenging [1, 2]. Although docking is widely used to generate starting structures for free-energy protocols, it usually underperforms in fragment-based drug discovery applications [3]. To address this limitation, we developed the Multiple-Copies Association Studies (MAS) workflow, an MD-based approach that can identify plausible binding modes without prior knowledge of the protein binding site.

MAS combines high-ligand-concentration simulations with a tailored Lennard-Jones adjustment to suppress ligand–ligand aggregation, followed by density-based clustering and lifetime analysis to prioritize stable and recurrent poses. In this study, we optimized and validated the MAS workflow using a diverse set of 23 protein–ligand complexes. Our results showed that MAS consistently identifies the correct ligand binding site and has the potential to detect the experimentally observed binding modes in more than half of the tested cases. The MAS validation studies also clarified the applicability domain of the approach, highlighting several factors that could influence the binding mode prediction process, including ligand protonation and tautomerization, conformational flexibility and ligand affinity.

Although not designed specifically for free-energy workflows, our results showed that MAS could be used as a complementary tool to support free-energy calculations by providing a set of plausible binding modes, particularly in challenging projects where experimental structural information is missing or incomplete. Furthermore, the MAS workflow could identify alternative, chemically reasonable binding orientations that may influence the predicted binding affinities and need to be considered before the free-energy calculations.

- 1) Cournia Z, Allen B, Sherman W. Relative Binding Free Energy Calculations in Drug Discovery: Recent Advances and Practical Considerations. *J Chem Inf Model.* 2017;57(12):2911-2937. doi:10.1021/acs.jcim.7b00564
- 2) Schindler CEM, Baumann H, Blum A, et al. Large-Scale Assessment of Binding Free Energy Calculations in Active Drug Discovery Projects. *J Chem Inf Model.* 2020;60(11):5457-5474. doi:10.1021/acs.jcim.0c00900
- 3) Lee TS, Allen BK, Giese TJ, et al. Alchemical Binding Free Energy Calculations in AMBER20: Advances and Best Practices for Drug Discovery. *J Chem Inf Model.* 2020;60(11):5595-5623. doi:10.1021/acs.jcim.0c00613

P53 - Optimizing ABFE Workflows for applications to membrane proteins GPCRs

Nithishwer Mouroug Anand, Antonia-Florina Panescu, Fabio Zuccotto and Philip Charles Biggin

Mouroug Anand

Accurate prediction of protein–ligand binding affinities is a central challenge in drug discovery. Among available computational approaches, Absolute Binding Free Energy (ABFE) calculations remain the gold standard due to their rigorous thermodynamic foundation. However, their application to membrane proteins, particularly G protein-coupled receptors (GPCRs), which constitute approximately 40% of current drug targets, remains insufficiently characterised.

Here, we systematically evaluate the performance of ABFE calculations across multiple GPCR systems, including the adenosine A2A receptor (A2A), dopamine D3 receptor (D3R), and metabotropic glutamate receptor 5 (mGlu5).

We identify three major sources of systematic error that limit ABFE performance in these systems: (i) inadequate resolution of the binding pocket in apo-like states, (ii) incomplete or inconsistent treatment of functionally important metal ions, and (iii) insufficient sampling of receptor conformational transitions between holo-like and apo-like ensembles.

To address these limitations, we introduce a set of methodological advances. We implement a nonequilibrium candidate Monte Carlo (NMC)-inspired water sampling strategy to enhance binding pocket resolution along the alchemical pathway, particularly in ligand-decoupled states. In addition, we develop a replica-exchange umbrella sampling (REUS)-based framework to quantify and correct for the free energy associated with receptor relaxation from holo-like to apo-like conformations.

These approaches significantly reduce systematic overestimation of binding affinities, improving agreement with experiment while preserving ranking performance. Our results highlight the critical role of apo-state representation in ABFE calculations and provide a practical framework for extending high-accuracy free energy methods to membrane protein targets.

P54 - Mining Druggable Sites in Influenza A Hemagglutinin: Binding of the Pinanamine-Based Inhibitor M090

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Influenza viruses cause approximately half a million deaths every year worldwide. In the search of novel therapeutic strategies, hemagglutinin (HA) has emerged as an attractive target for antiviral therapy due to its essential role in virus entry into the host cell. Several HA inhibitors have been explored to target either the HA-host cell interaction or prevent the pH-induced release of the fusion peptide that triggers the membrane fusion process, but with limited success. Assessing the binding mode of drug-like compounds is key in structure-based drug design. However, this may be challenged by factors such as the structural flexibility of the target protein. In this case, state-of-the-art computational methods can be valuable to explore the linkages between structural and pharmacological data. Following this strategy, extended molecular dynamics simulations and thermodynamic integration calculations are used to examine the binding of the antiviral inhibitor M090 and related pinanamine-based analogues covering a 250-fold difference in inhibitory potency to the influenza A HA, which is essential for virus entry and membrane fusion. This analysis has disclosed the hydrophobic shielding effect played by the 3-cyclopropylthiophene moiety in M090. Furthermore, the results support the negative effect of the resistance-induced E742→D mutation, which should weaken the binding by increasing the structural flexibility of the L2-BS loop. The results pave the way to explore the antiviral activity of novel compounds.

P55 - Optimising Potency Predictions: When and How FEP Data Improves Machine Learning Models

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Predicting the binding affinity of drug like molecules is essential in early drug discovery. Molecular Dynamics (MD) based methods such as Free Energy Perturbation (FEP) provide statistically rigorous estimates, while Machine Learning (ML) methods infer structure-activity relationships directly from data. Their performance, however, critically depends on having sufficient experimental potency measurements.

To supplement limited experimental data, here we systematically evaluate when augmented computational data from 3D sources (either from MD simulations or ML predictions) can meaningfully improve the accuracy of potency models. We first establish an experimental-only baseline and then assess its robustness by adding controlled Gaussian noise to emulate computational uncertainty and highlight inherent limitations of learning from noisy affinity estimates.

Building on the robustness analysis, we next investigate whether computationally derived affinity estimates can actively replace experimental measurements. Across multiple public and anonymised industry datasets, models trained with augmented data achieve predictive performance comparable to experimentally trained baselines. We further analyse how performance depends on (i) the amount of augmented data, (ii) molecular similarity and diversity, and (iii) the affinity range captured computationally. We establish degradation thresholds for both augmented-only and hybrid (augmented + experimental data) regimes, identifying when the model becomes inapplicable for prospective predictions. For selected compounds, we compare direct FEP predictions with ML based predictions trained on experimental versus augmented datasets, identifying cases where augmented modelling outperforms direct simulation.

Overall, our study provides quantitative guidelines for using computationally augmented data as a surrogate for experimental measurements when constructing potency models, clarifying when augmentation improves accuracy, when it introduces risk, and how to balance experimental and computational data to optimise predictive model performance in real-world drug discovery pipelines.

P56 - Decoding Isoform Selectivity in AMPK via Free Energy Calculations

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As a central regulator of cellular energy homeostasis, AMP-activated protein kinase (AMPK) represents a compelling therapeutic target for metabolic disorders such as type 2 diabetes [1]. However, the development of potent and selective modulators necessitates a rigorous understanding of their binding thermodynamics and molecular recognition mechanisms. In this study, we characterized the binding profiles of known allosteric pan-activators of AMPK [2] by employing molecular dynamics simulations combined with thermodynamic integration (TI) to calculate relative binding free energies [3]. Our computational results demonstrated excellent agreement with experimental data, identifying the key interactions within the Allosteric Drug and Metabolite (ADaM) site that drive ligand affinity. Through detailed thermodynamic analysis, we identified favorable substituent modifications capable of enhancing potency by up to 5 kcal mol⁻¹ and proposed novel scaffolds for future synthesis and biological evaluation. These findings underscore the efficacy of free energy calculations in rationalizing binding affinities and provide a robust foundation for the structure-based design of next-generation, isoform-selective AMPK therapeutics.

[1] Hardie DG, et al., *Annu Rev Nutr*, 34 (2014), 31–55.

[2] Ovens AJ, et al., *Biochem J*, 479 (2022), 1181–1204.

[3] Wang L, et al., *J Am Chem Soc*, 137 (2015), 2695–2703.

P57 - Folding Free Energy Perturbation Reveals a Critical Role of the PGLE Motif in Vreteno ZNF Stability

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Vreteno is a PIWI-associated protein central to the piRNA amplification cycle in Bombyx mori, potentially forming higher-order oligomers through interactions between its short and long (L) isoforms. Here we investigate the relative folding free energy of a Vreteno zinc finger (ZNF) dimer model (Vreteno L2) derived from AlphaFold-Multimer predictions, with particular focus on the conserved Pro–Gly–Leu–Glu (PGLE) motif.

Free energy perturbation (FEP) calculations were performed using a PMX 3.0 workflow with the AMBER99SB force field. A thermodynamic cycle was employed to compute changes in folding free energy upon mutation, using tripeptide reference systems to represent the unfolded state and a folded ZNF dimer model spanning residues 110-166 of the long isoform. Two mutation sites were selected: I137, located in the structured ZNF core, and L113 within the PGLE motif. Each site was mutated to alanine, glycine, valine, and threonine to probe effects of side chain size, hydrophobicity, and polarity.

The results reveal a pronounced site dependence of mutational effects. Mutations within the PGLE motif led to significantly larger destabilization (up to +25.5 kJ/mol in L113G) compared to mutations in the ZNF core (up to +11.2 kJ/mol for I137A). These findings indicate that the PGLE motif plays a critical role in maintaining ZNF dimer integrity within the L2 architecture.

Ongoing work aims to extend these calculations to binding free energies between the individual ZNF domains of the L2 dimer and to evaluate force field dependence by transitioning from AMBER99SB to CHARMM36. This work establishes alchemical free energy methods as a framework to quantify the energetic contribution of short linear motifs to the stability of multimeric protein domains.

P58 - Computational assay of hERG ion permeation, small-molecule modulation, and binding free energies

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The human ether-a-go-go-related gene (hERG) potassium channel is a critical antitarget in drug discovery, as off-target blockade can prolong the QT interval and trigger fatal arrhythmias.[1] Despite its pharmacological importance, atomistic simulations of hERG have been severely limited by the instability of its selectivity filter (SF), which prevents sustained ion permeation and restricts mechanistic studies of channel modulation.[2]

Here, we present a dual computational approach to characterise hERG at the atomistic level. First, combining the SF-stabilising N629D mutation with electronic continuum correction (ECC) force fields, we enable microsecond-timescale Computational Electrophysiology (CompEL) simulations of hERG under applied voltage. These simulations reproduce key functional properties, including inward rectification and single-channel conductance, and allow us to propose a structural basis for inward rectification. Simulations of classical inhibitors show that blockade arises through steric occlusion of the SF, whereas fenestration-bound activators enhance K⁺ conductance two- to threefold relative to apo.

Second, we apply non-equilibrium (NEQ) free energy calculations to calculate absolute binding free energies of hERG inhibitors. Preliminary results show promising agreement with experimental potency data, with further refinement ongoing to improve convergence. Together, these approaches establish a validated computational framework for studying hERG pharmacology, with direct applications in cardiotoxicity assessment and structure-guided drug design.

[1] M. Recanatini, E. Poluzzi, M. Masetti, A. Cavalli and F. De Ponti, *Med. Res. Rev.*, 2005, 25, 133–166.

[2] W. E. Miranda, K. R. DeMarco, J. Guo, H. J. Duff, I. Vorobyov, C. E. Clancy and S. Y. Noskov, *Proc. Natl. Acad. Sci. U. S. A.*, 2020, 117, 2795–2804.

P59 - N-Glycosylation Dynamics and Conformational Free Energy Landscapes in Full-Length IgG2 & IgG4 Monoclonal Antibodies: An All-Atom Molecular Dynamics Investigation

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Monoclonal antibodies (mAbs) represent a cornerstone of modern biotherapeutics and diagnostics, yet their conformational behaviour in solution remains incompletely characterised — particularly with respect to the role of post-translational modifications. N-glycosylation at the conserved Asn-297 sites of the Fc domain heavy chains is known to stabilise the Fc fragment and modulate Fcγ receptor engagement, but its distal influence on antibody-wide conformational dynamics has received limited attention.

Existing computational studies of N-glycosylation are predominantly restricted to isolated Fc fragments, or employ full-length models in which glycan chains are reconstructed from template crystal coordinates — an approach that introduces structural artefacts and may misrepresent the native glycan conformational ensemble. Crucially, the contribution of glycosylation to the free-energy landscape governing interdomain flexibility across the full antibody architecture — spanning Fc, hinge, and Fab regions — has not been systematically quantified.

In this work, we present all-atom molecular dynamics simulations of two full-length antibody structures (IgG2 and IgG4) taken directly from the PDB with their native, experimentally resolved glycan chains intact. Each system was simulated for 1.5 μs under explicit solvent conditions, providing sufficient sampling to characterise the conformational ensemble of both the glycans and the antibody framework. Free energy landscapes were constructed along collective variables capturing interdomain orientation and glycan conformational states, revealing how glycosylation reshapes the accessible conformational space — including at the Fab fragments, distal from the glycosylation sites themselves.

These results provide the first comprehensive, free-energy-resolved characterisation of N-glycosylation dynamics in the context of a complete antibody, with direct implications for the rational design and optimisation of glycoengineered therapeutic mAbs.

P60 - Cancer-associated mutations reconfigure dynamical responses in EGFR kinase

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The epidermal growth factor receptor (EGFR) is a tyrosine kinase receptor that plays a fundamental role in regulating cellular proliferation, survival, and differentiation. EGFR's kinase domain catalyses the autophosphorylation that drives downstream signalling and is a known hotspot for cancer-associated mutations. Such substitutions often lead to enhanced receptor activation, driving oncogenesis and therapeutic resistance, with major implications for prognosis and treatment choice; consequently, a detailed understanding of this domain's functional dynamics and the ways mutations reshape it is essential. Here, we apply the emerging dynamical-nonequilibrium molecular dynamics (D-NEMD) simulation method to probe how changes at the ATP-binding site are transmitted throughout the kinase domain, ultimately modulating the behaviour of distal regions in the wild-type protein and in two of the most clinically-observed non-small cell lung cancer variants, L858R and L858R+T790M.

D-NEMD, which combine both equilibrium and nonequilibrium simulations, uncovered previously uncharacterised communication networks linking the ATP pocket to key functional motifs, including the P-loop, α C-helix, hinge region, activation loop and α F-helix, exposing the intricate allosteric connectivity within this domain. Notably, numerous cancer-related mutation sites, despite being distant from the ATP site, map onto or lie close to these pathways, highlighting their functional relevance. Our simulations further revealed that the two investigated variants rewire EGFR's internal communication networks in distinct ways: L858R diminishes the transmission of structural changes to distal regions of the C-lobe, whereas introducing T790M on top of L858R partially restores wild-type-like behaviour through compensatory rerouting of early-stage dynamical responses.

P61 - Evaluating the impact of dummy atom corrections

Hannah Baumann, Irfan Alibay, Josh Horton, James Eastwood, Mike Henry, Alyssa Travitz, David Mobley

Open Free Energy, the Open Molecular Software Foundation

Alchemical free energy methods have found tremendous success over the last few decades, becoming key components of drug discovery pipelines. Despite many and ongoing innovations in this area, with several methodological improvements published each year, it remains challenging to consistently run free energy campaigns using state-of-the-art tools and best practices. In many cases, doing so requires expert knowledge, and/or the use of expensive closed source software. The Open Free Energy project (<https://openfree.energy/>) was created with the aim of addressing these issues. A joint effort between several academic and industry partners, the consortium aims to create and maintain reproducible and extensible open source tools for running large scale free energy campaigns.

In this presentation, we will cover recent developments that expand OpenFE's capabilities across workflows, protocols, and execution. Methodological updates include support for membrane systems within OpenFE protocols, broadening the range of applicable targets. We also introduce multi-method network strategies combining the hybrid topology protocol for closely related ligands with the separated topology protocol for larger or more complex transformations. This allows users to construct networks according to chemical similarity, increasing flexibility and efficiency. We will also highlight recent benchmark efforts evaluating the performance of the OpenFE tooling in production settings.

P62 - Estimator, the sampler, and the force field: These three are one

Yuanqing Wang

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The free energy differences among physical states of molecular systems offer a quantitative lens for understanding key biological processes, such as protein-ligand binding. The in silico estimation thereof, however, involves a multitude of heuristically designed procedures whose accuracy and efficiency have space for improvement—the determination of free energy differences, the traversing of the conformational landscape, and the curation of the force fields. With recent advances in graph representation, probabilistic inference, and control optimization, we show that it is possible to use machine learning, in a principled way, to enhance the modeling, sampling, and estimation stage, respectively. Finally, we offer a unified view of these aspects, and envision a blueprint for a foundation model jointly optimizing all three stages, towards a generalizable, efficient, and accurate characterization of Boltzmann distributions.

P63 - FEP protocols for real world potency optimisation

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The institute of cancer research

In recent years computational modelling has become a key part of both large- and small-scale drug discovery. Techniques such as docking and pharmacophore screening are widely used to identify new drug candidates and optimise potency, while the modelling of ADMET properties and solubility plays a vital role in understanding viability. Here we assess the role of FEP within this computational pipeline, as well as its deployment in a small-scale academic environment.

This assessment is posed as a series of key questions:

- Can the potency predictions provided by FEP calculations, applied in conjunction with generative models for molecular design, provide sufficient insight to guide better molecular design decisions?*
- How can accuracy and precision be balanced with throughput requirements? And can advanced sampling techniques be employed to aid in this balance?*
- Can the growing number of open-source solutions provide the tools required to deploy FEP at scale?*

This project aims to answer these questions by benchmarking a pair of open-source tools, namely openFE[1] and BioSimSpace[2], on an internal protein ligand dataset from a real drug discovery project. In addition to benchmarking these frameworks, we explore the potential of enhanced sampling methods such as replica exchange, REST and GCMC, and attempt to construct an optimal workflow that balances throughput and precision for real world applications.

[1] Baumann, H. M.; Horton, J. T.; Henry, M. M.; Travitz, A.; Ries, B.; Gowers, R. J.; Swenson, D. W. H.; Pulido, I.; Rufa, D.; Dotson, D. L. Large-Scale Collaborative Assessment of Binding Free Energy Calculations for Drug Discovery Using OpenFE. ChemRxiv 2025, DOI: 10.26434/chemrxiv-2025-7sthd

[2] Hedges et al., (2019). BioSimSpace: An interoperable Python framework for biomolecular simulation. Journal of Open Source Software, 4(43), 1831, <https://doi.org/10.21105/joss.01831>

P64 - Does the membrane matter? Assessing the explicit membrane protocol in OpenFE

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Membrane proteins account for about 70% of the druggable target space, yet accurately predicting the binding affinities of small molecules in these systems remains a significant computational challenge. The recent addition of explicit membrane support in OpenFE now enables users to set up alchemical transformations containing a protein-membrane system. However, while a lipid bilayer is an essential component of any physically complete model, it remains unclear to what extent current membrane setup parameters actually translate into improved predictive accuracy. Here, we present preliminary benchmarks of both Relative and Absolute Binding Free Energy (RBFE and ABFE) methods in OpenFE, varying key protocol parameters—such as lipid composition and system preparation choices—to assess their impact on predictive accuracy. We evaluate these protocols on various transporter and channel proteins grouped by how closely their binding sites are to the membrane. Given the local, short-timescale nature of alchemical perturbations, the influence of the lipid bilayer may be negligible for binding pockets distant from the membrane interface. By comparing these calculations with experimental affinities reported in recent literature, we investigate how the physical context of the binding pocket (e.g., membrane-exposed or solvent-exposed) dictates the need for a lipid environment, while noting that performance on soluble-like systems may not generalize to targets where the membrane is biologically essential. Ultimately, these insights aim to guide the community on when the added computational cost of having an explicit membrane is practically justified for alchemical simulations.

P65 - Fast Screens or Rigorous Ranking? AI Cofolding Models versus Free Energy Perturbation for GLP-1 Variant Binding to GLP1R

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Naturally occurring missense variants in glucagon-like peptide-1 (GLP-1) may contribute to inter-individual differences in incretin signaling and metabolic physiology. Using pharmacological data for 22 human GLP-1(7-37) missense variants identified across 500,000 UK Biobank participants, we compared AI-based predictive approaches (AlphaMissense, AlphaFold2 Multimer, AlphaFold3, Boltz-2, Chai-1, and a GLP1R-tailored MTNN) with relative binding free energy (RBFE) calculations for their ability to predict experimental binding affinities to the GLP-1 receptor (GLP1R).

RBFEs were calculated in GROMACS using pmx hybrid topologies and bidirectional nonequilibrium alchemical transitions in the amber99sb force field. The receptor-peptide complex was embedded in a POPC membrane, solvated with TIP3P water and 0.15 M NaCl. For each mutation, three replicas were run for states A and B, with 100 transition trajectories initiated from equilibrium snapshots at each endpoint, yielding 300 forward and 300 reverse transitions per mutation, per leg.

FEP showed the strongest agreement with experiment when charge-shifting substitutions were excluded (Pearson $r = -0.80$; Spearman $\rho = -0.88$). When all variants were included, AlphaFold2 Multimer's ipTM+ptm metric gave the highest overall correlation ($r = 0.65$; $\rho = 0.67$), while the GLP1R-specific MTNN provided a more robust ranking of variant effects.

These results highlight both the promise and limitations of current AI-based cofolding and predictive tools for peptide-receptor design. AI models enable rapid, low-cost screening and can help distinguish poor from promising binders, especially when tailored to a specific target. However, they do not replace rigorous free-energy methods for quantitative prioritization. Instead, our results support a tiered workflow in which AI is used for fast screening and FEP for final prioritization before synthesis and pharmacological characterization.

P66 - Computational alchemy for studying protein covalent modifications

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Many biological processes involve covalent modification of amino acid side chains. Here, we extend our physics-based alchemical free-energy framework to predict thermodynamic changes from side-chain (de)protonation, disulfide bond formation, and post-translational modifications. We demonstrate quantitative accuracy across these processes and highlight applications to pH-dependent gating of membrane proteins, insulin stability, and the impact of post-translational modifications on amyloid fibril stability.

P67 - How many crystal structures do you need to trust your docking results?

Alexander Matthew Payne, Benjamin Kaminow, Hugo MacDermott-Opeskin, Iván Pulido, Jenke Scheen, Maria A Castellanos, Daren Fearon, John D. Chodera, Sukrit Singh

Memorial Sloan Kettering Cancer Center

Structure-based drug discovery technologies generally require the prediction of putative bound poses of protein:small molecule complexes to prioritize them for synthesis. The predicted structures are used for a variety of downstream tasks such as pose-scoring functions or as a starting point for binding free energy estimation. The accuracy of downstream models depends on how well predicted poses match experimentally-validated poses. Although the ideal input to these downstream tasks would be experimental structures, the time and cost required to collect new experimental structures for synthesized compounds makes obtaining this structure for every input intractable. Thus, leveraging available structural data is required to efficiently extrapolate new designs. Using data from the open science COVID Moonshot project—where nearly every compound synthesized was crystallographically screened—we assess several popular strategies for generating docked poses in a structure-enabled discovery program using both retrospective and prospective analyses. We explore the tradeoff between the cost of obtaining crystal structures and the utility for accurately predicting poses of newly designed molecules. We find that a simple strategy using molecular similarity to identify relevant structures for template-guided docking is successful in predicting poses for the SARS-CoV-2 main viral protease. Further efficiency analysis suggests template-based docking of a scaffold series is a robust strategy even when the quantity of available structural data is limited. The resulting open source pipeline and curated datasets should prove useful for automated modeling of bound poses for downstream scoring, machine learning, and free energy calculation tasks for structure-based drug discovery programs.

P68 - An AI Agent for Streamlining Binding Free Energy Calculations

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Genentech

Predicting binding free energy is a central challenge in structure-based drug design. Molecular dynamics (MD)-based approaches, particularly free energy perturbation (FEP) methods, are increasingly adopted in drug discovery due to advances in algorithms, GPU acceleration, and the availability of open-source frameworks such as OpenFE. However, FEP workflows remain complex, involving multiple steps and requiring substantial domain expertise—factors that limit broader adoption among computational and medicinal chemists.

To address these challenges, we developed an AI agent that abstracts workflow complexity and enables users to perform OpenFE-based binding free energy calculations through a conversational interface. This agent encapsulates best practices and automates key steps, lowering the barrier to entry while maintaining flexibility for expert users. Importantly, the agent supports customization, allowing users to tailor workflows to specific systems or project needs without requiring deep technical knowledge.

We further demonstrate how this agentic approach facilitates rapid dissemination and iteration of computational protocols. Unlike traditional software tools, the agent can be easily updated and shared, enabling continuous improvement without the need for extensive user retraining.

Case studies from large-scale deployment of OpenFE workflows highlight improvements in usability, efficiency, and reproducibility. More broadly, this work illustrates how AI agents can serve as an effective interface layer for complex computational workflows in drug discovery.

P69 - From Binding Thermodynamics to Free Energy Prediction: Molecular Recognition in Carbonic Anhydrase–Sulfonamide Model Systems

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Accurate prediction of protein–ligand binding thermodynamics remains a major challenge in computational drug design, especially in systems where experimentally measured binding parameters are affected by binding-linked protonation of both the protein and the ligand. In such cases, experimentally observed binding affinities do not correspond directly to intrinsic binding affinities, because intrinsic parameters are obtained by correcting for contributions arising from these linked protonation effects. This distinction is especially important when experimental data are used to benchmark computational methods intended to describe the fundamental energetic determinants of binding. In this study, carbonic anhydrase–sulfonamide complexes are used as well-defined model systems to investigate molecular recognition and its relationship to binding thermodynamics in the context of free energy prediction. These systems are particularly suitable because sulfonamide binding to the catalytic zinc site is strongly influenced by protonation state, while high-quality structural and thermodynamic data are available. To support this analysis, we use data from the Protein–Ligand Binding Database (PLBD; <https://plbd.org>), which provides curated observed and intrinsic binding parameters together with structural information for protein–ligand complexes. Using these datasets and corresponding crystal structures, we compare computational methods against both observed and intrinsic thermodynamic references. Various structure-based drug design (SBDD) methods are used to predict binding affinities, while ligand-based drug design (LBDD) approaches, including QSAR modeling with diverse molecular descriptors and multiple machine learning algorithms, are also employed.

Our results indicate that structure-based approaches show improved agreement with intrinsic rather than observed binding parameters in systems affected by binding-linked protonation. In contrast, QSAR models show similar predictive performance for observed and intrinsic datasets, although calibration against intrinsic binding parameters may provide added value for interpretation and mechanistic analysis. Overall, this study highlights the importance of rigorous thermodynamic reference data for improving free energy prediction in drug design.

P70 - Nonequilibrium Alchemical Free Energy Calculations for Protein Mutations Using OpenFE: Application to ABL1 Binding Systems

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Accurate prediction of how protein mutations affect ligand binding is critical for understanding drug resistance, protein engineering, and molecular recognition. Relative binding free energy (RBFE) calculations offer a rigorous framework for estimating these effects, yet their application to protein mutations remains challenging compared to more established ligand transformation workflows.

We present an implementation and validation of nonequilibrium (NEQ) cycling alchemical free energy calculations for protein mutations within the OpenFE framework. By extending OpenFE's existing infrastructure for ligand transformations, we integrate protein mutation support into a consistent, reproducible, and interoperable workflow built on the OpenMM simulation engine. This unified approach streamlines setup, execution, and analysis of protein mutation RBFE calculations, lowering the barrier for adoption and enabling systematic benchmarking across diverse systems.

We validate our approach on the ABL1 kinase system, computing relative binding free energies for a set of experimentally characterized mutations in complex with imatinib. Our results demonstrate good agreement with experimental measurements, supporting the reliability of the NEQ protocol for protein mutations.

We further explore the ABL1 + ATP system, which has proven challenging for standard free energy protocols. We investigate the effect of lambda scheduling strategies within the NEQ framework as a route to improving prediction accuracy for difficult transformations. Insights into how tailored lambda schedules may address convergence challenges in such systems will be discussed.

This work contributes to the growing toolkit for alchemical free energy calculations for protein mutations and highlights the flexibility of the OpenFE ecosystem for method development and benchmarking.

P71 - More protein-ligand data are needed for AlphaFold-like models to enable drug discovery

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Structure-based drug design (SBDD) is an evolving paradigm that leverages protein structural information to improve small molecule therapeutic design. Building on more than 50 years of data curation from the Protein Data Bank, the recent emergence of protein structure prediction models (PSPMs) promises to enable new computationally driven approaches for therapeutic discovery. However, it is critical to assess the limitations of these models using blind challenges and to expand existing datasets to better reflect real-world drug design tasks. Here, we discuss recent efforts to benchmark existing PSPMs and identify their limitations. We offer a hierarchical framework for parsing which tasks the current models perform well, and which tasks remain challenging or unexplored. Finally, we emphasize the need for systematic dataset generation to support the development of frontier models and highlight recent efforts to generate experimental and physics-based datasets for challenging tasks in drug discovery.

P72 - Accurate in silico prediction of pH-dependent antibody binding affinities using robust physics-based methods

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Schrödinger GmbH

The ability to modulate antibody (Ab) behavior with respect to pH has broad implications in biologics design. However, predicting these pH-dependent binding affinities is challenging for standard empirical models. Here we introduce two physics-based methodologies: Constant-pH Molecular Dynamics (CpHMD) to identify pH-sensing residues, and Protein Free Energy Perturbation (Protein FEP) to calculate binding free energy changes with robust treatment of alternate protonation states. Validated against experimental data in two distinct case studies, these in silico tools accurately replicate pH-dependent binding to accelerate biologics design.

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